

BasTo6809 Compiler

User Manual and Language Reference

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About this manual

This edition reorganizes the original BasTo6809 documentation into a maintainable reference guide. Command syntax, examples, memory-layout tables, hardware notes, and example programs are retained from the original manual while typography, navigation, terminology, and page structure are standardized.

Note: Compatibility note: Some commands require a specific Color Computer model, CoCoSDC, CoCoMP3, or compiler option. Check each command entry and the applicable hardware section before use.

Document conventions

- Command signatures appear in a shaded monospaced heading.
- Examples appear in monospaced code blocks and use compiler-ready straight quotation marks.
- Hexadecimal values use the compiler's &H prefix or the 6809 assembler's \$ prefix, depending on context.
- A # placeholder denotes a numeric argument unless the command entry states otherwise.

Quick Start

1. Unzip the BASIC_To_6809 package on your computer.
2. Open a terminal and change to the new BASIC_To_6809 folder.
3. On macOS, you may need to remove the downloaded-file quarantine before the included executables will run. From the BASIC_To_6809 folder, enter: `xattr -d com.apple.quarantine *`
`2>/dev/null`. Windows users can skip this step.
4. Start the Super Duper Extended Color BASIC IDE with `./SDECB`. On Windows, use `SDECB.exe`.
5. Enter a BASIC program in the IDE or load `TEST.BAS` from the File menu. The IDE checks syntax and helps find typing errors before compilation.
6. Choose a compile command from the Compile menu, or press F11 for the default build configured in the makefile (`compile.bat` on Windows).
7. The default build compiles the BASIC source to 6809 assembly, assembles it with `lwasm`, creates `DISK1.DSK` from `BLANK.DSK`, and copies the generated program to the disk image with `decb`.
8. Run the disk image in an emulator or copy it to an SD card for use on a real CoCo with a CoCoSDC. The makefile and `compile.bat` are intended to be customized for your workflow.

Once `TEST.BIN` builds successfully, you are ready to compile your own BASIC programs. Have fun!

Command Line Options

BasTo6809 accepts the options below. Options placed in a `CompileOptions` comment at the start of a `.BAS` file take precedence over command-line options. Example:

EXAMPLE

```
'CompileOptions -i -notext -r80 -s32 -o2 -fArcade_B0_F1
```



-COCO

Read a tokenized Color Computer BASIC input file.

-ascii

Read a plain-text ASCII BASIC input file. This is the normal choice for files created in a text editor or the SDECB IDE.

-dragon

Generate output for a Dragon computer.

-mX

Select floating-point precision: `-m0` uses the faster 3-byte format; `-m1` uses the more accurate 5-byte format.

-i

Use strict integer-only compilation. Default numeric variables are `INTEGER`, integer division truncates toward zero, and floating-point literals, types, operators, or functions produce compile-time errors. Floating-point libraries are omitted. `LONG`, `_INTEGER64`, and unsigned integer types remain available and include only the integer support they require.

-SXXX

Set the maximum size of ordinary strings, from 1 through 255 bytes. The default is 255. Fixed-length declarations such as DIM NAME\$ AS STRING * 32 can set a smaller maximum for an individual string; assignments are truncated to the declared maximum.

-o0 / -o1 / -o2

Select the optimization level. The default is -o2; -o0 disables optimization.

-notext

Do not reserve the standard \$0400-\$05FF text screen. The low-memory layout begins at \$0400 instead of \$0600. This option cannot be combined with the CoCo 3 WIDTH command.

-pxxxx

Set the compiled program's start address to hexadecimal xxxx. The address must be at or above the safe address calculated from resident loaders, shared RAM, and graphics pages. The compiler reports an error instead of silently moving an unsafe address.

-fxxxx

Select the graphics-screen font. The default is Arcade_B0_F1. Available fonts are in Basic_Includes/GraphicsMode/Graphic_Screen_Fonts.

-a

Make the generated program autostart after loading.

-r

Return to BASIC when the compiled program ends instead of entering the normal endless loop.

-Vx

Set compiler verbosity. -V0 produces minimal output; higher values provide progressively more detail.

-v

Display the BasTo6809 version number.

-k

Keep miscellaneous intermediate files generated during compilation.

-h

Display command-line help.

-rXXXX

Reserve xxxx hexadecimal bytes of shared RAM between resident loaders and graphics memory. The area is left untouched so CHAINED programs can share variables, pointers, tables, or other data. Chained programs must use a compatible layout.

Resident loaders

-rxxxx shared RAM

Alignment to the next \$0200 boundary

Graphics memory

Alignment to the next \$0100 boundary

Compiled program

All values supplied to -rxxxx are hexadecimal.



No reserved RAM

Options and commands	Loader locations	Graphics start	Program start without graphics
Default, no CHAIN	—	\$600	\$600
-notext, no CHAIN	—	\$400	\$400
SDC_CHAIN	SDC: \$0600–\$07A5	\$800	\$800
-notext, SDC_CHAIN	SDC: \$0400–\$05A5	\$600	\$600
Disk CHAIN	Disk: \$0600–\$09EF	\$0A00	\$0A00
-notext, disk CHAIN	Disk: \$0400–\$07EF	\$800	\$800
Both loaders	SDC: \$0600–\$07A5; Disk: \$07A6–\$0B95	\$0C00	\$0C00
-notext, both loaders	SDC: \$0400–\$05A5; Disk: \$05A6–\$0995	\$0A00	\$0A00

With -r80

-r80 reserves \$0080, or 128 bytes.



Options and commands	Loaders end	Reserved shared RAM	Graphics start	Program start without graphics
-r80, no CHAIN	\$600	\$0600-\$067F	\$800	\$800
-notext -r80, no CHAIN	\$400	\$0400-\$047F	\$600	\$600
-r80, SDC_CHAIN	\$07A6	\$07A6-\$0825	\$0A00	\$0A00
-notext -r80, SDC_CHAIN	\$05A6	\$05A6-\$0625	\$800	\$800
-r80, disk CHAIN	\$09F0	\$09F0-\$0A6F	\$0C00	\$0C00
-notext -r80, disk CHAIN	\$07F0	\$07F0-\$086F	\$0A00	\$0A00
-r80, both loaders	\$0B96	\$0B96-\$0C15	\$0E00	\$0E00
-notext -r80, both loaders	\$996	\$0996-\$0A15	\$0C00	\$0C00

GMODE 16 examples

One GMODE 16 page consumes \$1800 bytes.



For an ordinary program, `GMODE 16,0` reserves one page and `GMODE 16,1` reserves pages 0 and 1. The largest literal page named anywhere in the source determines the number of CoCo 1/2 graphics pages reserved by the compiler.

`SDC_PLAYMOVIE` is a special case on the CoCo 1/2. Its player always uses two \$1800-byte framebuffers, so the compiler reserves \$3000 bytes even when the source only contains `GMODE 16,0`. A later CHAINED program is compiled independently. If that target does not use `SDC_PLAYMOVIE` and only selects `GMODE 16,0`, it reserves one page and may place its program in the former second movie page. This reclaims the memory; it also destroys the old movie frame, as expected when CHAIN replaces the first program.

Configuration	Reserved shared RAM	Graphics memory	Automatic program start
Default, GMODE 16	—	\$0600–\$1DFF	\$1E00
-notext, GMODE 16	—	\$0400–\$1BFF	\$1C00
-notext -r40, GMODE 16	\$0400–\$043F	\$0600–\$1DFF	\$1E00
-notext -r80, GMODE 16	\$0400–\$047F	\$0600–\$1DFF	\$1E00
-notext, both loaders, GMODE 16	—	\$0A00–\$21FF	\$2200
-notext -r80, both loaders, GMODE 16	\$0996–\$0A15	\$0C00–\$23FF	\$2400
-r80, both loaders, GMODE 16	\$0B96–\$0C15	\$0E00–\$25FF	\$2600

The generated assembly lists `CHAIN_SHARED_START`, `CHAIN_SHARED_SIZE`, `CHAIN_SHARED_END`, `GRAPHICS_PAGE_START`, and `PROGRAM_SAFE_START`. Compare these symbols when checking the compatibility of programs in a chain. Changing the GMODE page count can lower `PROGRAM_SAFE_START`; it does not by itself move or remove the `-rxxxx` shared area.

The reason `-notext -r40` moves the graphics start from \$0400 to \$0600 is:

```

$0400–$043F   Reserved shared RAM
$0440–$05FF   Alignment space
$0600         Next valid $0200-aligned graphics address

```

An explicit `-pxxxx` may place the program at a higher address, but it cannot overlap the loaders, shared area, or graphics pages.

If using CoCo 3 WIDTH command, you cannot use the `-notext` option:

```

WIDTH 80 uses $1220 bytes:
80 columns × 29 rows × 2 bytes = 4640 bytes = $1220

```

The layout order is:

```

CHAIN loaders

```

`-rxxxx` shared RAM

Align to \$0200

WIDTH 80 screen

Align to \$0100

Compiled program

WIDTH 80 without -rxxxx

BASIC commands used	Loader memory	WIDTH 80 screen	Program start
No CHAIN loader	—	\$0600–\$181F	\$1900
SDC_CHAIN	\$0600–\$07A5	\$0800–\$1A1F	\$1B00
Disk CHAIN	\$0600–\$09EF	\$0A00–\$1C1F	\$1D00
Both loaders	\$0600–\$0B95	\$0C00–\$1E1F	\$1F00

WIDTH 80 with -r80

-r80 reserves \$80, or 128 bytes.



BASIC commands used	Loader memory	Shared -r80 RAM	WIDTH 80 screen	Program start
No CHAIN loader	—	\$0600–\$067F	\$0800–\$1A1F	\$1B00
SDC_CHAIN	\$0600–\$07A5	\$07A6–\$0825	\$0A00–\$1C1F	\$1D00
Disk CHAIN	\$0600–\$09EF	\$09F0–\$0A6F	\$0C00–\$1E1F	\$1F00
Both loaders	\$0600–\$0B95	\$0B96–\$0C15	\$0E00–\$201F	\$2100

For any -rxxxx value:

Shared start = first byte after the loaders

Shared end = shared start + \$xxxx - 1

Screen start = next \$0200 boundary after shared memory

Program start = next \$0100 boundary after screen start + \$1220

For example, with both loaders and -r200:



Region	Address
SDC loader	\$0600-\$07A5
Disk loader	\$07A6-\$0B95
Shared RAM	\$0B96-\$0D95
Alignment	\$0D96-\$0DFF
WIDTH 80 screen	\$0E00-\$201F
Program	\$2100 onward

`-notext` cannot be combined with `WIDTH 80` or other `WIDTH` commands; that combination produces a compile-time error.



Enhancements to Colour BASIC

- Write BASIC programs using the helpful included Super Duper Extended Colour Basic (SDECB) IDE.
- Using the OPTION _EXPLICIT feature in the SDECB IDE it will show you unassigned variables, making it easy to catch typos.
- New GMODE command allows you to choose every graphic mode the CoCo can produce, including semi graphics and if using a CoCo 3 all of the CoCo 3 graphics modes and Colour modes. Using these new screens you can use LINE (with B & BF), CIRCLE and PAINT commands.
 - Can print directly to any graphic screen using PRINT #-3,
 - Use of line numbers is optional
 - You can use Labels for sections of code to jump to (case sensitive)
 - Variable names can be 25 characters long (case sensitive)
- Doesn't use any ROM calls (Extended BASIC ROM not required), possible to use all of the 64k of RAM on Coco 1 & 2
 - Added new sprite commands
- Added commands for the CoCo 3 to use scrollable playfields (backgrounds)
 - Added new audio sample commands
- Many new SDC related commands allow you to read and write directly to the SDC filesystem from your BASIC program
- A new SDC audio streaming command to play RAW music files/audio samples directly from the SD card in the CoCoSDC
- Easily insert your own assembly code anywhere you want in your program and easily share values of variables between BASIC and your assembly code.
- Added keyword Const which let's you label a value and use the value in your program, but doesn't take any RAM in your final compiled program.
- The assembly language code generated is fully commented showing each BASIC line and how it's compiled. The assembly file generated can be used to help someone learn how to program in assembly language. Or allow an experienced assembly programmer to optimize the program by hand.

Changes to BASIC's Graphic features

PMODE has been replaced by the GMODE command PCLS has been replaced by the GCLS command PCOPY has been replaced with the GCOPY command LINE command format has been changed to include a colour value and no longer uses PSET and PRESET.

The commands PSET, HSET, PRESET, HRESET, PPOINT or HPOINT are not supported. Instead they are replaced with SET and POINT commands. The compiler will use whichever graphic mode is set using the GMODE command and will SET pixels to the requested colour the user wants that matches the GMODE requested.

You can now use SET,POINT,LIN,CIRCLE & PAINT commands on every screen, even the regular text screen, using GMODE 0,1

GMODE ModeNumber,GraphicsPage Selects the graphics screen and the graphics page. ModeNumber is the graphics mode you want to use GraphicsPage is the Page you want to show/use for your graphics commands To see a list of ModeNumbers and the resolutions go [here](#) Special note the ModeNumber must be an actual number and cannot be a variable as the compiler needs to know exactly which graphic mode commands to be included at compile time. GraphicsPage can be a variable. If you are going to use Graphic pages, the compiler needs to know how many pages to reserve in RAM (for CoCo 1 & 2 graphics). So you must have a GMODE #,MaxPages entry at the beginning of your BASIC program. Where the value of MaxPages will be an actual number and not a variable.

```
GCLS #
```



Colour the graphics screen # is the colour value you want the screen to be coloured

```
GCOPY SourcePage,DestinationPage
```

Makes a copy the Source graphics page to the Destination graphics page. SourcePage - Source graphics page DestinationPage - Destination graphics page

```
SET(x,y,Colour)
```



Sets a pixel on the screen x,y - Screen location of the pixel to be drawn

Colour - Colour Number of the pixel to be drawn

POINT(x,y) Returns the colour value of the pixel selected x,y - Screen location of the pixel value requested

```
LINE(x0,y0)-(x1,y1),Colour[,B][F]
```



x0,y0 - Starting location x1,y1 - Ending location Colour - Colour of the Line or Box to draw B - Draw a Box F - Fill the Box

```
PAINT(x,y),OldColour,FillColour
```



Fills the old colour value with the fill colour value which must also be the border colour of the section you are painting x,y - Starting location OldColour - Colour Number FillColour - Colour Number

```
CIRCLE(x,y),Radius,Colour
```



Draws a circle on the screen x,y - Origin of the circle Radius - Size of the circle, to keep the aspect ratio close to round Some of the graphics modes use scaling so the Radius isn't always a count of actual pixel values. Colour - Colour Number of the circle to draw

```
PALETTE v,Colour
```



Sets the CoCo 3 Palette value

v -palette slot of 0 to 15

Colour - Colour value of 0 to 63

GET & PUT commands are not available

New commands and features added to Color BASIC

- IF/THEN/ELSE/ELSEIF/ENDIF
- SELECT/CASE
- WHILE/WEND
- DO/WHILE/LOOP
- DO/LOOP/UNTIL
- COCOHARDWARE a Command that let's you identify which CoCo your program is running on
V=COCOHARDWARE(0)
- COCOMP3 Commands See here for more info

– SDC_PLAY Command that plays an audio sample or song directly off the SD card in the SDC Controller. See here for more info

– SDC_PLAYORCL, SDC_PLAYORCR, SDC_PLAYORCS these commands are similar to SDCPLAY except the audio is sent to the Orchestra 90 or COCOFLASH cartridge. See here for more info

– SDC file access commands that allow you to Read & Write files directly on the SD card's own filesystem. See here for more info

– GETJOYD - Quickly get the joystick values of 0,31,63 of both joysticks both horizontally and vertically

- PRINT #-3, - Print to the graphics screen created with the GMODE command. Can also be used as ?#-3,"Hello World!". The default font is ArcadeArcade_B0_F1, but you can select others using the compiler command -fxxxx Where xxxx is the font name used for printing to the graphics screen (default is Arcade_B0_F1). Look in folder Basic_Includes/GraphicsMode/Graphic_Screen_Fonts to see font names available
- SLEEP Used to pause program execution
- PLAYFIELD # - Used to set the Scrollable playfield mode. The # given must be an actual number and not a variable. # Max Resolution Min Resolution Size Multiple 1 - 256 x 7872 256 x 192 not applicable x 64 2 - 512 x 3840 512 x 192 not applicable x 192 3 - 1024 x 512 1024 x 256 not applicable x 256 4 - 2048 x 256 512 x 256 256 x not applicable 5 - 2560 x 192 512 x 192 256 x not applicable
- VIEW x,y - Command to select where in the scrollable playfield you want to see. The x & y coordinates are the top left corner of a viewable window of the playfield.

SPRITE_LOAD "SpriteName.asm", #[,f] — Load a Sprite The **SPRITE_LOAD** command loads a sprite into your program. • "SpriteName.asm" is the name of your sprite file (must be in assembly format). Generated by the command line tool called PNGtoCCSprite • # is the sprite number — a numeric ID used to reference this sprite in your program when using actual **SPRITE** commands. • f is the number of frames this sprite has can be left blank if the sprite has only one frame. This command associates the loaded sprite with the given number, so you can easily manage multiple sprites in your game.

SPRITE Command — Control Sprites on the Screen The **SPRITE** command, along with various options, allows you to control the appearance and behaviour of sprites in your program.

Command Reference:

```
SPRITE OFF [#]
```

Turns off sprite number #.

```
If no number is provided, all sprites are turned off.  
SPRITE LOCATE #, x, y
```

Moves sprite # to a new position on the screen. x and y are the playfield coordinates.

```
SPRITE SHOW #[, f]
```

Displays sprite # using frame f.

```
If the sprite is not animated (i.e. it has only one frame) use of ,f is optional.  
SPRITE BACKUP #
```

Saves the background behind sprite #. This allows the sprite to be cleanly erased later.

```
SPRITE ERASE #
```

Erases sprite # and restores the background previously saved with **SPRITE BACK**.

```
SPRITE #,x,y[, f]
```

Draws sprite number # at x,y co-ordinates using frame f. f is optional, a value of 0 (single frame sprite) is used if not given.

WAIT VBL - Wait for Vertical Blank then update the sprites on screen

TRIM\$(string) Removes spaces from both ends of a string string - String variable

LTRIM\$(string) Removes spaces from the beginning of a string string - String variable

RTRIM\$(string) Removes spaces from the end of a string string - String variable

Adding Assembly code to your program

You can insert your own assembly code directly in your BASIC program anywhere you like. To insert your own assembly code into the program you can do so by using the special words "ADDASSEM:" and "ENDASSEM:" after a REM (') symbol and a space as:

```
' ADDASSEM:
```

```
LDA    #$FF
RORB
```

```
' ENDASSEM:
```

Another example of putting your own assembly code in the BASIC program:

```
Print "HEY"
GoSub Mycode
Print "HELLO"
...
Mycode:
' ADDASSEM:
    LDA    #$FF
    STA    $400
' ENDASSEM:
Return
```

Accessing Numeric variables in your assembly code

You access numeric variables by adding the prefix "Var" to the name. For example if you have an integer variable in your program called Counter and you want to read the value of the variable you could use code like this:

```
DIM Counter As Integer
PRINT "COUNTER=";Counter
' ADDASSEM:
    LDD     _Var_Counter    ; Get the variable in D
    ANDB    #%11111110     ; Make the number even
    STD     _Var_Counter    ; Save the updated value
' ENDASSEM:
PRINT "COUNTER=";Counter
```

Accessing String variables in your assembly code

You access string variables by adding the prefix "StrVar" to the name. Strings are stored as data where the first byte is the length of the string the rest of the bytes are the actual string data.

For example if you want to interact with a string variable called A\$ in assembly you could use code like this:

```
' ADDASSEM:
    LDX    #_StrVar_A      ; X points at the first byte in the A$ variable
    LDB    ,X+             ; B = string length and X points at its first character
; First byte is the length of the string, from 0 to 255
    BEQ    @Finished      ; If the string length is zero then skip past
    LDU    #$400           ; Text screen start
!   LDA    ,X+             ; Get a byte from the string
    STA    ,U+             ; Show the byte on screen
    DECB                   ; Decrement the counter
    BNE    <              ; Keep looping until B=0
@Finished:                 ; @ in the label makes it a local label
; Always leave a blank line after code that uses a local @ label
' ENDASSEM:
```

Accessing Arrays in your assembly code

You access string variables by adding the prefix "*StrVar*" to the name. Strings are stored as data where the first byte is the length of the string the rest of the bytes are the actual string data.

For example if you want to interact with a string variable called A\$ in assembly you could use code like this:

```
Dim Buffer(1024) As _Unsigned _Byte
Dim BufferStart As _Unsigned Integer
10 GoSub DoStuff
DoStuff:
' ADDASSEM:
    LDX    #_ArrayNum_Buffer
    STX    _Var_BufferStart
' ENDASSEM:
Return
```

Compiling from the command line

It is highly recommended that you use the SDECB IDE as it will show you syntax errors and other errors that will help to compile your program without errors. You can also compile your BASIC programs to machine language without the IDE, from the command line using the following:

```
Using MacOS or Linux:
./BasTo6809 -ascii BASIC.bas
lwasml -9bl -p cd -o./ML.bin BASIC.asm > ./Assembly_Listing.txt
Using Windows:
.\BasTo6809.exe -ascii BASIC.bas
lwasml -9bl -p cd -o./ML.bin BASIC.asm > ./Assembly_Listing.txt
```

At this point you'll have an EXECutable program called ML.bin in the folder that you can use on a real CoCo or an emulator.

Optimizing

The 6809 CPU has the hardware MUL instruction and it is used for many of the multiplication routines even the large 16, 32, 64 bit integer and the floating point math routines use it. Division is always going to be slower as it is all software based. Therefore if you want to maximize the speed of your program, try to use multiplication instead of division if possible.

For example if you have something like this:



```
A = B / 2
```

Use:



```
A = B * .5
```

If variables are not assigned a type using the DIM command they will default to Single (Fast Floating Point). To maximize the speed of the 6809 CPU try to use 8 bit (fastest) or 16 bit types (fast). Other variable types will take more space and will run slower. See the next page for all the variable types.

Variable Types

Normally, a numeric variable whose type cannot be inferred defaults to SINGLE. With -i, it defaults to INTEGER and floating-point use is rejected at compile time. It's best if you assign variables with the DIM statement or use the symbol values as a suffix after your variable name or literal number, to use the minimum size needed.

Type	Suffix	Declaration	Range
1	`	_Bit	Min -1, Max 0
2	~`	_Unsigned _Bit	Min 0 , Max 1
3	%%	_Byte	Min -128, Max 127
4	~%%	_Unsigned _Byte	Min 0, Max 255
5	%	Integer	Min -32,768, Max 32,767
6	~%	_Unsigned Integer	Min 0, Max 65,535
7	&	Long	Min -2 Gig, +2 Gig
8	~&	_Unsigned Long	Min 0, Max 4 Gig
9	&&	_Integer64	Min -(8 byte value), Max +(8 byte value)
10	~&&	_Unsigned _Integer64	Min 0, Max (8 byte value)
11	!(None)	Single (Default size)	Min E-20, Max E+19 (Fast Float)
12	#	Double	Min E-308, Max E+308

*** Types 7 to 12 will slow your program down considerably.

Examples of assigning variable types:

```
DIM MyVar1 AS _Byte  
DIM MyVar2 AS _UNSIGNED LONG  
DIM MyVar3 AS DOUBLE
```



Generally you don't need to force a variable to a different type then assigned, the compiler will scale the types between each other. But this is an example of forcing the type:

A = V~%% ' Force V to be an Unsigned Byte

64k programs

Use `cc1sl` (CoCo 1 Super Loader) when the first CoCo program requires more than 32K. Also use it when Disk BASIC starts a compiler-generated program containing `CHAIN` or `SDC_CHAIN`, even if the main program is small: the resident chain loader occupies low RAM used by Disk BASIC while loading. `cc1sl` creates a startup file that safely handles both cases, including programs that use addresses normally occupied by BASIC ROM.

Dragon programs compiled with `-dragon` do not require this step. `DRAGLOAD.BIN` handles 64K Dragon programs directly.

cc1sl usage

```
cc1sl [-l] [-vx] FILENAME.BIN -oOUTNAME.BIN [.scn | .csv]
```

- `-l` displays `LOADING` at the bottom of the screen while the program loads.
- `-vx` selects verbosity 0, 1, or 2. The default is `-v0`.
- `FILENAME.BIN` is the assembled CoCo machine-language program.
- `-oOUTNAME.BIN` sets the output filename; the default is `GO.BIN`.
- An optional `.scn` file supplies a binary CoCo text screen displayed while loading.
- An optional `.csv` file supplies a CSV text screen displayed while loading.

See `cc1sl_help.txt` for the complete utility reference.

The steps for compiling a big program are:

```
For MacOS and Linux:
./BasTo6809 HELLO.BAS
lwasm -9bl -p cd -o./HELLO.BIN HELLO.asm > ./NEW_Assembly_Listing.txt
./cc1sl -l HELLO.BIN -oBIGFILE.BIN
For Windows:
.\BasTo6809 HELLO.BAS
lwasm -9bl -p cd -o./HELLO.BIN HELLO.asm > ./NEW_Assembly_Listing.txt
.\cc1sl -l HELLO.BIN -oBIGFILE.BIN
```

In this case your final program to execute on the CoCo is called `BIGFILE.BIN`, you can of course call it whatever you want.

Preparing programs for CHAIN and SDC_CHAIN

`CHAIN` and `SDC_CHAIN` replace the running program with another compiled DECB-format machine-language program. Both resident chain loaders can load large, multi-segment programs throughout the 64K address space. This changes where `cc1sl` belongs in a chain-loading project:

Program in the sequence	File to use	Run through <code>cc1sl</code> ?
First compiler-generated program containing <code>CHAIN</code> or <code>SDC_CHAIN</code> , initially loaded from Disk BASIC	The <code>cc1sl</code> output	Yes
First program without a resident chain loader and entirely within the normal Disk BASIC loadable area	Normal assembled <code>.BIN</code>	No
Program loaded by <code>CHAIN</code>	Normal assembled <code>.BIN</code>	No
Program loaded by <code>SDC_CHAIN</code>	Normal assembled <code>.BIN</code>	No

Only the initially launched program may use the `cc1sl` wrapper. When Disk BASIC launches a compiler-generated program containing a resident chain loader, wrap that first program even if its main code is small. Never run a target of `CHAIN` or `SDC_CHAIN` through `cc1sl`. The chain command is itself the 64K loader; it expects the compiler's normal DECB LOADM file and loads every segment before jumping to the execution address in that file's postamble.

Both chain loaders force the computer into all-RAM mode before copying the target, so the target may contain segments and executable code above `$7FFF` even if the calling program entered `CHAIN` while the BASIC ROMs were selected. The floppy loader temporarily selects ROM mode only during physical disk-controller transfers so the CoCo 1/2 hardware NMI path remains available; it returns to all-RAM mode before processing the loaded bytes.

Example build sequence

Suppose `START.BAS` launches `PART2.BIN`:

```
CHAIN "PART2.BIN:0"
```

Compile and assemble both programs normally. Use `cc1sl` only on the first program:

```
./BasTo6809 START.BAS
lwasm -9bl -p cd -oSTART.BIN START.asm
./cc1sl -l START.BIN -oGO.BIN

./BasTo6809 PART2.BAS
lwasm -9bl -p cd -oPART2.BIN PART2.asm
```

Load and run `GO.BIN` from Disk BASIC. `GO.BIN` starts the first program; its `CHAIN` command then loads the unwrapped `PART2.BIN` and transfers control to it. The same rule applies when the first program uses:

```
SDC_CHAIN "PART2.BIN",0
```

In that case, copy the normal unwrapped `PART2.BIN` to the CoCoSDC filesystem.

Preparing every chained program

- Compile every program in the sequence with the same `-notext` setting. This keeps the low-memory base compatible.
- A program only includes the resident loader for the chain commands found in its source. If `PART2.BIN` will later execute `CHAIN` or `SDC_CHAIN`, include that command so the compiler adds the required loader.
- Programs that exchange data through `-rxxxx` shared RAM must use the same `-rxxxx` size, the same `-notext` setting, and the same resident-loader set (`CHAIN`, `SDC_CHAIN`, or both). These choices determine `CHAIN_SHARED_START`. The programs must also agree on the byte layout and meaning of the shared data.
- If a chain target must share parameters but will not itself execute the loader command, an unreachable reference can retain the required loader and compatible shared address. For example, `IF 0 THEN SDC_CHAIN "UNUSED.BIN",0` includes the SDC loader but is never executed.
- Ordinary BASIC variables are initialized by each new program and are not chain parameters. Store values that must survive in the reserved shared area before executing `CHAIN` or `SDC_CHAIN`.
- A chain target must be a valid DECB LOADM `.BIN` with an execution address in its postamble. Do not use the `cc1sl` output as a chain target.
- `CHAIN` and `SDC_CHAIN` do not return. Save any state needed by the next program before executing the command.

Graphics allocation is independent of the shared area. On a CoCo 1/2, `SDC_PLAYMOVIE` reserves two \$1800-byte frame pages in the program that uses it. A later program without movie playback can select only `GMODE 16,0`, reserve one page, and reuse the former second framebuffer without losing access to a correctly matched `-rxxxx` parameter area.

Compiling for a Dragon computer

There are a few things you need to do in order to compile and run a program on a Dragon computer. Since the Dragon computer is very similar to a CoCo 1 code from the compiler that works on a CoCo 1 & 2 will pretty much execute as is on a Dragon computer. But the keyboard layout is different between the two computers. When compiling your program to target a Dragon computer you must use the -dragon option. This tells the compiler to include the proper keyboard routines that will work for the Dragon computer. Also since the Dragon uses a different file format for loading binary files you must copy the DRAGLOAD.BIN file to your floppy disk or emulator disk image. You will also need to rename your compiled program to COMPILED.BIN and copy COMPILED.BIN to the same floppy disk or emulator disk image as DRAGLOAD.BIN. You can rename DRAGLOAD.BIN

The steps you would take to compile a program called HELLO.BAS

```
1) Compile your program into .asm file
./BasTo6809 -ascii -dragon HELLO.BAS
2) Assemble the program into a DECB machine language binary file
lwasm -9b -p cd -o./HELLO.bin HELLO.asm
3) Copy the DRAGLOAD.BIN file to a floppy disk or disk image and rename the file as
HELLO.BIN
./decb copy -2 -b -r ./DRAGLOAD.BIN DISK1.DSK, HELLO.BIN
4) Copy your actual program HELLO.bin to a floppy disk or disk image, and rename the
file as COMPILED.BIN
./decb copy -2 -b -r ./HELLO.bin DISK1.DSK,COMPILED.BIN
```

Your disk should now have two files on it called HELLO.BIN (which is really the DRAGLOAD.BIN program) and COMPILED.BIN (which is your actual compiled program). The loader expects the file to be loaded from drive 1 and you start your program on the Dragon computer by typing:
RUN"HELLO.BIN"

The DRAGLOAD.BIN program will handle programs that are 64k.

BASIC commands

ATTR foreground,background[,B][,U]

Sets the character attributes used by subsequent output on the CoCo 3 40-, 64-, and 80-column text screens.

- `foreground` - Foreground colour number from 0 through 7. These values use hardware palette registers 8 through 15.
- `background` - Background colour number from 0 through 7. These values use hardware palette registers 0 through 7.
- `B` - Optional blinking-text attribute.
- `U` - Optional underline attribute. `B`, `U`, both, or neither may be used.

When changing a foreground colour with `PALETTE`, add 8 to the `ATTR` foreground value to obtain its hardware palette-register number.

EXAMPLE

```
ATTR 3,2,U
```



AUDIO ON|OFF

Controls whether the cassette input is connected to the display speaker.

- `ON` - Route cassette-player audio to the speaker.
- `OFF` - Disconnect cassette audio and restore the normal DAC sound path used by commands such as `SOUND` and `PLAY`.

EXAMPLE

```
AUDIO OFF
```



CIRCLE (x,y),radius,colour

Draws a circle on the currently selected GMODE graphics page. The meaning of the colour value depends on that graphics mode. Some modes scale one axis to keep circles visually round, so `radius` is not always a literal pixel count.

- `x,y` - Centre of the circle.
- `radius` - Size of the circle.
- `colour` - Colour number used to draw the circle.

EXAMPLE

```
GMODE 16,0  
CIRCLE (128,96),40,3
```



CLEAR

Resets compiled numeric variables and numeric-array elements to zero, and empties strings and string-array elements. Legacy Color BASIC operands may be accepted, but string-space and highest-memory settings are ignored because the compiled program's memory layout is fixed at compile time.

EXAMPLE

```
CLEAR
```



CLS [c]

Clears the current text screen.

- On the 32-column screen, `c` may be a colour code from 0 through 8. With no argument, the screen uses its standard default fill.
- On CoCo 3 40-, 64-, and 80-column screens, `c` is a background palette index from 0 through 7. With no argument, palette index 0 is used.

EXAMPLE

```
CLS 2
```



CMP

On a CoCo 3, writes the standard composite-monitor colours into all 16 hardware palette registers.

EXAMPLE

```
CMP
```



COLOR c1,c2

Sets the default foreground and background colour values used by subsequent graphics commands. It does not recolour pixels that have already been drawn. Valid colour numbers depend on the active GMODE; `GCLS` without an argument uses the current background value.

- `c1` - Foreground colour.
- `c2` - Background colour.

EXAMPLE

```
COLOR 2,3
```



CONST name = value

Defines one case-insensitive compile-time constant. The tokenizer substitutes the value wherever the name is used, so the constant requires no runtime storage. Use a separate `CONST` statement for each name. Numeric expressions and quoted string values are supported; a string constant's name ends in `$`.

EXAMPLE

```
CONST ScreenWidth = 256  
CONST Title$ = "WEATHER"
```



COPYBLOCKS source,destination,count

Copies consecutive physical 8 KiB MMU blocks on a CoCo 3. This is intended for fast transfers such as double-buffered background screens. The block numbers are physical MMU blocks, not GMODE page numbers.

- `source` - First source block to be copied from
- `destination` - First destination block to be copied to
- `count` - Number of consecutive blocks to copy, from 1 through 255.

The caller must select blocks present in installed RAM and must avoid overwriting the running program or other live data. Do not pass zero: the runtime uses an unsigned byte counter, so zero would wrap and attempt 256 copies.

EXAMPLE

```
COPYBLOCKS &H10,&H20,3
```



CPUSPEED [#]

The compiler normally runs the CPU at the maximum speed supported by the detected hardware. Use `CPUSPEED` when a program needs a specific clock rate, for example when timing code for original hardware.

- `# = 1` - Select 6809-emulation mode at approximately 0.895 MHz.
- `# = 2` - Select 6809-emulation mode at approximately 1.79 MHz.
- `# = 3` - Select 6809-emulation mode at approximately 2.864 MHz on hardware that supports the high-speed mode.
- Omitted or any other value - Select the maximum supported hardware speed. An HD6309 is also placed in native mode.

EXAMPLE

```
CPUSPEED 1
```



This example selects approximately 0.895 MHz.

CoCoSDC file-transfer commands temporarily select the speed required by the controller and restore the speed selected by the program when they return. This includes `SDC_LOADM` and `SDC_SAVEM`; calling either command does not cancel an earlier `CPUSPEED` selection.

DATA constant,constant,...

Stores numeric and string constants for sequential retrieval by `READ`. Strings containing spaces or commas must be quoted. `RESTORE` resets the read position to the first item.

- `constant` - A numeric value or quoted string value.

EXAMPLE

```
DATA 45,"CAT",98
READ Score,Animal$,NextScore
```

DIM variable[(bounds)] [AS type]

Declares scalar variables or arrays and optionally selects their data types. Array bounds are inclusive upper bounds, so `Scores(9)` contains elements 0 through 9. Ordinary numeric variables default to `SINGLE` unless integer-only compilation is selected. See [Variable Types](#) for the complete type and range table.

A scalar string can have a fixed maximum length from 1 through 255 using `AS STRING * length`. Assignments longer than that maximum are truncated.

EXAMPLE

```
DIM Count AS _UNSIGNED INTEGER
DIM Scores(9) AS _BYTE
DIM Names$(3,10)
DIM Title$ AS STRING * 32
```

DO ... LOOP

Repeats a block of statements. A condition on `DO` is tested before the body; a condition on `LOOP` is tested after the body. Therefore a post-test loop runs at least once.

Supported forms are `DO ... LOOP`, `DO WHILE test ... LOOP`, `DO UNTIL test ... LOOP`, `DO ... LOOP WHILE test`, and `DO ... LOOP UNTIL test`. Use `EXIT DO` to leave the innermost DO loop early.

EXAMPLE

```
Do While A < 10
  A = A + 1
Loop

Do Until A > 9
  A = A + 1
Loop

Do
  A = A + 1
Loop Until A = 10

Do
  A = A + 1
Loop While A < 10
```

DRAW string

Interprets a Color BASIC draw string on the currently selected GMODE page. Commands may be separated with semicolons.

Draw command	Meaning
A0 through A3	Select an angle/rotation.
B	Move without drawing the next movement command.
Cn	Select drawing colour n .
Dn , En , Fn , Gn , Hn , Ln , Rn , Un	Move down, diagonally, left, right, or up by n units.
Mx,y	Move the drawing position to x,y .
N	Do not update the current position after the next movement.
Sn	Select scale; 4 is normal size.

If a direction command omits its distance, one unit is used.

EXAMPLE

```
GMODE 16,0  
SCREEN 1,0  
DRAW "BM128,96;U25;R25;D25;L25"
```



END

Terminates program execution immediately.

EXAMPLE

```
END
```



EXEC address

Jumps to the specified 16-bit machine-code address. `EXEC` does not establish a BASIC return address; the machine-code routine is responsible for transferring control somewhere appropriate.

EXAMPLE

```
EXEC &H7022
```



EXIT DO / EXIT FOR / EXIT WHILE

Exits the innermost matching `DO...LOOP`, `FOR...NEXT`, or `WHILE...WEND` loop immediately. Execution continues with the first statement after that loop's ending keyword.

EXAMPLE

```
WHILE A < 100  
  IF A = 10 THEN  
    EXIT WHILE  
  ELSE  
    A = A + 1  
  END IF  
WEND
```



FOR variable = start TO finish [STEP increment]

Begins a loop ending at `NEXT`. The finishing value is inclusive. `STEP` defaults to 1; use a negative increment to count downward.

- `variable` - Loop counter.
- `start` - Initial counter value.
- `finish` - Inclusive final value.
- `increment` - Optional amount added after each iteration.

EXAMPLE

```
FOR Z=35 TO 125 STEP 5
  PRINT Z
NEXT Z
```



GCLS [c]

Fills the currently selected GMODE graphics page. If `c` is omitted, the background value previously selected by `COLOR` is used. Valid colours depend on the active GMODE.

EXAMPLE

```
GCLS 2
```



GCOPY SourcePage, DestinationPage

Copies one complete graphics page in the active GMODE. Source and destination are zero-based graphics-page indexes, and both pages must have been reserved by the program.

- `SourcePage` - Source graphics page.
- `DestinationPage` - Destination graphics page.

EXAMPLE

```
GMODE 15,1
GMODE 15,0
GCLS 0
GCOPY 0,1
```



GETJOYD

Quickly reads both joysticks and quantizes each axis to 0, 31, or 63. Results are stored in the standard Color BASIC joystick locations:

Address	Value
<code>\$015A</code>	Left joystick vertical
<code>\$015B</code>	Left joystick horizontal
<code>\$015C</code>	Right joystick vertical
<code>\$015D</code>	Right joystick horizontal

EXAMPLE

```
GETJOYD
LeftVertical = PEEK(&H015A)
LeftHorizontal = PEEK(&H015B)
RightVertical = PEEK(&H015C)
RightHorizontal = PEEK(&H015D)
```



GMODE mode[,page]

Selects the graphics mode and the zero-based graphics page used by subsequent drawing commands. `GMODE ,page` changes only the page while retaining the current mode. `GMODE` selects the drawing page but does not display it; use `SCREEN 1,colourSet` to show graphics.

- `mode` must be a literal number because the compiler selects graphics libraries at compile time. See the [CoCo 1/2 GMODE table](#) and [CoCo 3 GMODE table](#).

- `page` may be an expression or variable. If omitted, page 0 is selected.

For CoCo 1/2 graphics, the largest literal page used determines how much RAM the compiler reserves. It is the highest zero-based page number: page 3 requires pages 0 through 3.

EXAMPLE

```
GMODE 16,0  
SCREEN 1,0
```

For the special NTSC composite 256-colour modes 160 through 165, the first such `GMODE` automatically initializes palette entries 0 through 3 to 0, 16, 32, and 48.

GOSUB line-or-label

Calls a subroutine at a line number or label. `RETURN` resumes execution at the statement following the `GOSUB`.

EXAMPLE

```
GOSUB ShowMessage  
PRINT "BACK FROM SUBROUTINE"  
END  
  
ShowMessage:  
    PRINT "IN SUBROUTINE"  
    RETURN
```

GOTO line or label

Jumps to the specified line number or label.

EXAMPLE

```
GOTO Complete  
PRINT "THIS IS SKIPPED"  
Complete:  
    PRINT "DONE"
```

IF test THEN ... ELSEIF ... ELSE ... END IF

Conditionally executes statements. Zero is false and a nonzero value is true. The compiler accepts both a single-line form, `IF condition THEN statement [ELSE statement]`, and the multiline form shown below. `ELSEIF` and `ELSE` are optional.

EXAMPLE

```
IF X=0 THEN
    PRINT "ZERO"
ELSEIF X<0 THEN
    PRINT "NEG"
ELSE
    PRINT "POS"
END IF
```



INPUT ["prompt";] variable[,variable...]

Reads keyboard input into one or more variables. An optional quoted prompt may precede the variable list. Enter multiple values separated by commas; each value is converted to the destination variable's declared type.

EXAMPLE

```
INPUT "NAME,AGE"; Name$,Age
```



LET variable = expression

Assigns a value to a variable (optional keyword).

EXAMPLE

```
LET A3=27
```



LINE (x0,y0)-(x1,y1),colour[,B|BF]

Draws a line or rectangle on the current GMODE page. The continuation form `LINE -(x1,y1),colour` begins at the previous LINE endpoint.

- `x0,y0` - Starting location.
- `x1,y1` - Ending location.
- `colour` - Drawing colour; valid values depend on the active GMODE.
- `B` - Draw an unfilled rectangle.
- `BF` - Draw a filled rectangle. `F` is not valid by itself.

EXAMPLE

```
LINE (22,33)-(27,39),1,BF
LINE -(60,50),2
```



LOOP

Ends a `DO` loop. A trailing `WHILE` test repeats while the condition is true; a trailing `UNTIL` test repeats until the condition becomes true. With no condition, `LOOP` repeats unconditionally. See `DO...LOOP` for every supported pre-test and post-test form.

EXAMPLE

```
DO
    INPUT "TYPE 0 TO FINISH"; Choice
LOOP UNTIL Choice = 0
```



LOCATE x,y

Sets the graphics-font cursor used by subsequent graphics-screen output such as `PRINT #-3, .`. Coordinates are character positions for the font selected by the compiler's `-f` option.

EXAMPLE

```
LOCATE 20,12
PRINT #-3,"HELLO"
```



LPOKE location,value

Stores an unsigned byte from 0 through 255 in CoCo 3 extended/banked memory. `location` may range from 0 through 2,097,151 (`&H1FFFF`). Use ordinary `POKE` for a 16-bit address in the current 64 KiB address space.

EXAMPLE

```
LPOKE 480126,241
```



MOTOR ON|OFF

Turns the cassette motor relay on or off.

EXAMPLE

```
MOTOR ON
```



NEXT v1,v2,...

Ends the current `FOR` loop. Optional variable names can close multiple nested loops and must be listed from the innermost loop to the outermost. If omitted, `NEXT` closes only the most recent `FOR`.

EXAMPLE

```
FOR X = 0 TO 3
  FOR Y = 0 TO 2
    PRINT X;Y
  NEXT Y,X
```



NTSC_FONTCOLOURS b,f

Sets the background and foreground values used by the special composite-output fonts in GMODE 160 through 165. Both values are unsigned bytes from 0 through 255.

- `b` - Background composite colour.
- `f` - Foreground composite colour.

EXAMPLE

```
NTSC_FONTCOLOURS 0,3
```



ON expr GOSUB line1,line2,...

Calls one of several subroutines using a one-based selector: 1 chooses the first destination, 2 chooses the second, and so on. Zero or a value beyond the list continues with the following statement. The selected subroutine returns normally with `RETURN`.

EXAMPLE

```
ON A GOSUB 100,230,500,1125
```



ON expr GOTO line1,line2,...

Branches using a one-based selector: 1 chooses the first destination, 2 chooses the second, and so on. Zero or a value beyond the list continues with the following statement.

EXAMPLE

```
ON A GOTO 100,230,500,1125
```



PAINT(x,y),OldColour,FillColour

Flood-fills the contiguous region of `OldColour` connected to `(x,y)`, replacing it with `FillColour` on the current GMODE page.

- `x,y` - A point inside the region.
- `OldColour` - Existing colour to replace.
- `FillColour` - New colour.

EXAMPLE

```
PAINT (44,55),2,3
```



PALETTE *pr,c*

Writes colour code *c* (0 through 63) to CoCo 3 palette register *pr* (0 through 15) immediately, and updates the compiler runtime's mirror entry.

EXAMPLE

```
PALETTE 1,13
```



PALETTEW *pr,c*

Writes colour code *c* (0 through 63) only to mirrored palette register *pr* (0 through 15). Hardware colours do not change until `PALETTEV` is used, so several entries can be prepared without visible intermediate changes.

EXAMPLE

```
PALETTEW 1,13
```



PALETTEV *pr,c*

Writes colour code *c* (0 through 63) to mirrored palette register *pr* (0 through 15), waits for vertical blank, and then copies all 16 mirrored entries to the hardware palette. This applies a prepared palette without visible tearing.

EXAMPLE

```
PALETTEW 0,0  
PALETTEW 1,13  
PALETTEV 2,26
```



PLAY *string*

Plays a semicolon-separated music command string.

Command	Meaning
A through G	Musical note.
N1 through N12	Select a note by number.
01 through 05	Select octave.
L	Set note length.
T	Set tempo.
V	Set volume.
P	Pause.
# or +	Sharp.
-	Flat.
.	Extend the preceding note.

EXAMPLE

```
PLAY "L1;A;A#;A-"
```



PLAYFIELD

Selects one of the CoCo 3 scrollable-playfield layouts. The mode must be a literal number from 1 through 5 because the compiler configures memory and includes support code at compile time.

Use `PNGtoCC3Playfield` to convert a PNG into the required binary file or files, then load those files with `LOADM`, `SDC_LOADM`, or `SDC_BIGLOADM` before using `VIEW`.

Mode	Minimum resolution	Maximum resolution	Width multiple	Height multiple
1	256 x 192	256 x 7872	Fixed at 256	64
2	512 x 192	512 x 3840	Fixed at 512	192
3	1024 x 256	1024 x 512	Fixed at 1024	256
4	512 x 256	2048 x 256	256	Fixed at 256
5	512 x 192	2560 x 192	256	Fixed at 192

EXAMPLE

```
PLAYFIELD 2
' Load the converted playfield files here
VIEW 0,0
```



POKE location,value

Stores a value (0–255) in a memory location (0–65535 decimal or 0–\$FFFF hexadecimal).

EXAMPLE

```
POKE 28000,241
```



PRINT [#device,] expression-list

Prints numeric or string expressions. Semicolons concatenate fields and suppress the final newline; commas advance to print zones.

On `WIDTH 40`, `WIDTH 64`, and `WIDTH 80` screens, each comma advances to the next 16-column print zone and wraps or scrolls at the screen edge. These wide text operations use the active relocated framebuffer selected by the compiler's memory layout.

- With no device, output goes to the current text screen.
- `#-2` sends output through the CoCo serial/printer path.

- #-3 through #-5 select graphics-screen output as described below.

EXAMPLE

```
PRINT "SCORE:";Score  
PRINT #-2,"PRINTER TEST"
```



PRINT #-3,expression

Prints on the currently selected GMODE graphics page. The default font is `Arcade_B0_F1`; a compiler option such as `-fArcade_B0_F1` selects a font from

`Basic_Includes/GraphicsMode/Graphic_Screen_Fonts`. Use `LOCATE` to set the graphics-font cursor.

EXAMPLE

```
GMODE 16,0  
LOCATE 2,3  
PRINT #-3,"HELLO"
```



PRINT #-4,expression

Prints normal 42-column by 24-row text on GMODE 16 using a dedicated 6x8 one-bit font: black background with white foreground.

EXAMPLE

```
PRINT #-4,"NORMAL"
```



PRINT #-5,expression

Prints inverse 42-column by 24-row text on GMODE 16 using the same 6x8 font: white background with black foreground.

EXAMPLE

```
PRINT #-5,"INVERSE"
```



PRINT TAB(n)

Moves the cursor to column `n` on the low- and high-resolution text screens.

EXAMPLE

```
PRINT TAB(22);"HELLO!"
```



PRINT@position,message

Prints a message at zero-based character position 0 through 511 on the 32-column by 16-row text screen. Position 0 is the upper-left corner and position 511 is the lower-right corner.

EXAMPLE

```
PRINT@11,"HELLO!"
```



READ var1,var2,...

Reads successive values from `DATA` statements into the listed variables. The compiler performs no end-of-DATA check; reading too many values advances into unrelated memory and produces undefined results. Use `RESTORE` to reset the read pointer.

EXAMPLE

```
READ A1,B,C7
```



REM comment

Inserts a remark (comment). BASIC ignores everything after REM on that line.

EXAMPLE

```
REM THIS IS A COMMENT LINE
```



RESTORE

Resets the READ pointer back to the first item on the first DATA line.

EXAMPLE

```
RESTORE
```



RETURN

Returns from a subroutine to the statement following the last GOSUB.

EXAMPLE

```
RETURN
```



RGB

On a CoCo 3, writes the standard RGB-monitor colours into all 16 hardware palette registers.

EXAMPLE

```
RGB
```



RUN

Restarts the current compiled program at its generated `START` entry point. The runtime is initialized again, variables are cleared, and the `DATA` pointer is reset.

EXAMPLE

```
RUN
```



SAMPLE SINGLE slot | LOOP slot | OFF

Controls CoCo 3 audio-sample playback. A slot must first be populated by `SAMPLE_LOAD`.

- `SINGLE slot` - Play the selected slot once.
- `LOOP slot` - Repeat the selected slot continuously.
- `OFF` - Stop sample playback.

Sample slots range from 0 through 31.

EXAMPLE

```
SAMPLE SINGLE 2
```



SAMPLE_LOAD "filename",slot

Loads a raw unsigned 8-bit, 6003 Hz audio file into CoCo 3 memory and assigns it a slot from 0 through 31 for use by `SAMPLE`. The filename and slot must be compile-time literals, and the file must exist when the program is compiled.

EXAMPLE

```
SAMPLE_LOAD "JUMP.raw", 0
```



SCREEN [type][,colourSet]

Selects text or graphics display and optionally selects the VDG colour set. The compiler accepts `SCREEN type`, `SCREEN type,colourSet`, and `SCREEN ,colourSet`.

- `type = 0` - Display the text screen.
- Any nonzero `type` - Display the graphics mode and page selected by `GMODE`.
- `colourSet = 0` - Select colour set 0.
- Any nonzero `colourSet` - Select colour set 1.

EXAMPLE

```
SCREEN 0,1
```



SELECT CASE expr ... CASE ... CASE ELSE ... END SELECT

Provides a multi-way branch based on `expr`. BASIC executes the first matching `CASE` block. If none match, it executes `CASE ELSE` when that optional block is present.

EXAMPLE

```
Select Case A
  Case 1
    Print "ONE"
  Case 2
    Print "TWO"
  Case Else
    Print "OTHER"
End Select
```

SET(x,y,Colour)

Sets one pixel on the current graphics page. Coordinate and colour limits depend on the active GMODE.

- `x,y` - Pixel coordinates.
- `Colour` - Pixel colour.

EXAMPLE

```
SET (11,11,3)
```

SLEEP milliseconds

Pauses program execution for approximately the requested number of milliseconds, from 0 through 65,535. Timing is based on HSYNC and is quantized in roughly 10 ms steps; values below 21 currently produce no delay.

- `milliseconds` - Requested pause duration.

EXAMPLE

```
SLEEP 1000
```

SOUND tone,duration

Produces a tone through the DAC.

- `tone` - Pitch value from 1 through 255.
- `duration` - Duration value from 1 through 255, using Color BASIC-compatible timing units.

EXAMPLE

```
SOUND 33,22
```



SPRITE option

Queues sprite operations for the double-buffered graphics display. Sprite slots range from 0 through 31. `WAIT VBL` processes queued operations during vertical blank and swaps the buffers so changes become visible.

Form	Operation
<code>SPRITE OFF [slot]</code>	Disable one sprite, or all sprites when the slot is omitted.
<code>SPRITE LOCATE slot,x,y</code>	Set a sprite's playfield coordinates.
<code>SPRITE SHOW slot[,frame]</code>	Queue a sprite for display.
<code>SPRITE BACKUP slot</code>	Save the background behind a sprite.
<code>SPRITE ERASE slot</code>	Restore the background saved by <code>BACKUP</code> .
<code>SPRITE slot,x,y[,frame]</code>	Draw a sprite directly at the supplied coordinates.

Animated frame indexes start at 0. The frame argument is optional for a single-frame sprite.

EXAMPLE

```
SPRITE LOCATE 1,80,60  
SPRITE SHOW 1,2  
WAIT VBL
```



SPRITE_LOAD "SpriteName.asm", # [,f]

Loads a sprite definition from an assembly-format file generated by `PNGtoCCSB`, assigning it to a slot from 0 through 31. The filename, slot, and optional frame count must be compile-time literals, and the file must exist at compile time. If `f` is the total frame count, runtime frame indexes range from 0 through `f-1`.

Sprites are supported in every GMODE, but the generated file must match the program's GMODE because memory layout, palette, coordinate limits, and aspect ratio vary by mode.

EXAMPLE

```
SPRITE_LOAD "HERO.asm", 1, 4
```



STOP

Halts the compiled program in an endless loop. It does not return to a BASIC interpreter.

EXAMPLE

```
STOP
```



TIMER = value

Sets the unsigned 16-bit runtime timer to `value` (0 through 65,535). The timer increments once per 60 Hz interrupt and wraps from 65,535 to zero. Read the current value with the numeric `TIMER` function.

EXAMPLE

```
TIMER = 120
```



UNTIL

Used with `DO` or `LOOP` to repeat until a condition becomes true. `DO UNTIL test` is a pre-test form and may run zero times; `LOOP UNTIL test` is a post-test form and always runs the body at least once.

EXAMPLE

```
D0 UNTIL Finished
    GOSUB ProcessNextItem
LOOP
```



VIEW x,y

Selects the visible viewport of a previously configured `PLAYFIELD`. `x` and `y` give the top-left playfield coordinate. Valid ranges depend on the selected playfield mode and converted image dimensions.

EXAMPLE

```
VIEW 0,0
```



WAIT VBL

Waits for vertical blank, processes queued sprite backup/draw/erase operations, and swaps the double buffers. Use it after configuring sprites with `SPRITE`.

EXAMPLE

```
WAIT VBL
```



WHILE test ... WEND

Repeats a block of statements while `test` is true. When it becomes false, execution continues with the statement after `WEND`. `WHILE` can also be used in the pre-test and post-test forms of `D0...LOOP` described under `D0`.

EXAMPLE

```
While A < 10
    A = A + 1
Wend
```



WIDTH columns

Selects the text-screen resolution. `columns` must be one of these literal values:

Value	Text resolution	Hardware
32	32 x 16	CoCo 1, 2, or 3
40	40 x 28	CoCo 3
64	64 x 28	CoCo 3
80	80 x 28	CoCo 3

Wide modes reserve screen RAM and cannot be combined with the `-notext` compiler option.

EXAMPLE

```
WIDTH 80
```



Disk Related Commands

These commands access standard DECB floppy disks. Drive suffixes use :0 through :3; drive 0 is used when the suffix is omitted. The sequential interface supports one reader and one writer at the same time when they use different file numbers.

```
x = _FILEEXISTS(string)
```

Returns 255 (-1) when the named floppy-disk file exists, or 0 when it does not. The filename may include an optional drive suffix.

EXAMPLE

```
A = _FILEEXISTS("DATA.DAT:1")
```

```
CHAIN "FILENAME.BIN[:drive]"
```

Loads a DECB machine-language program from floppy disk, replaces the current program, and executes the address in the LOADM postamble. The drive number is optional. The .BIN extension is added when no extension is supplied. CHAIN does not return to the original program.

The target must be the normal .BIN produced by assembling the compiler's output. Do not process a program that will be loaded by CHAIN with cc1s1; CHAIN already contains the loader needed to load a large program across the 64K address space. A compiler-generated first program containing CHAIN should be processed with cc1s1 when Disk BASIC loads it, even if its main code is small, because the resident loader occupies part of Disk BASIC's loading workspace. See [Preparing programs for CHAIN and SDC_CHAIN](#) for the complete build procedure.

EXAMPLE

```
CHAIN "PART2.BIN:1"
```

```
OPEN "FILENAME.EXT[:drive]","X",#
```

Opens a DECB disk file for sequential byte access. Use R to read an existing file or W to create or replace a file. # is file number 0 or 1. A reader and writer may be open together, but they must use different file numbers.

EXAMPLE

```
OPEN "DATA.DAT:2","R",0
```



CLOSE(#)

Closes file number 0 or 1. Closing an output file flushes buffered data and commits the FAT and directory updates. An output file must contain at least one byte before it is closed. The disk motor is turned off automatically after disk activity stops.

EXAMPLE

```
CLOSE(0)
```



PUTBYTE0 x

Writes the low eight bits of x to output file 0. The file position advances automatically by one byte.

EXAMPLE

```
PUTBYTE0 65
```



PUTBYTE1 x

Writes the low eight bits of x to output file 1. The file position advances automatically by one byte.

EXAMPLE

```
PUTBYTE1 ASC("A")
```



x = GETBYTE(#)

Reads and returns one byte from input file 0 or 1. The file position advances automatically. GETBYTE returns 0 after end-of-file; because zero is also a valid byte, use LOF to control loops that must read every byte.

EXAMPLE

```
B = GETBYTE(0)
```



x = LOF(#)

Returns the length of open file 0 or 1 as an unsigned 32-bit value (`_UNSIIGNED LONG`). Input files report their complete length; output files report the bytes written so far, including buffered bytes.

EXAMPLE

```
FileSize = LOF(0)
```



SETPOS0(x)

Sets file 0 to zero-based unsigned 32-bit byte offset x. For input, this only changes the read position and never writes data; seeking beyond end-of-file stops at the end. For output, seeking forward fills skipped bytes with zeros. Seeking backward in an output file produces a runtime error.

EXAMPLE

```
SETPOS0(299)
```



SETPOS1(x)

Sets file 1 to zero-based unsigned 32-bit byte offset x. For input, this only changes the read position and never writes data; seeking beyond end-of-file stops at the end. For output, seeking forward fills skipped bytes with zeros. Seeking backward in an output file produces a runtime error.

EXAMPLE

```
SETPOS1(0)
```



A\$ = FILEINFO\$(#)

Returns a 32-byte record for open file 0 or 1. Bytes 1-8 contain the space-padded name; 9-11 the extension; byte 12 the attributes (zero for DECB); 13-28 are reserved; and 29-32 contain the unsigned 32-bit size, least-significant byte first. With fixed-length strings, A\$ must hold at least 32 bytes.

EXAMPLE

```
A$ = FILEINFO$(0)
```



x = DELETE(A\$)

Deletes the named DECB file. A\$ may include a drive suffix. No sequential input or output file may be open. Returns 0 when deleted and 5 when not found; other disk failures produce a runtime disk error.

EXAMPLE

```
R = DELETE("OLD.DAT:1")
```



x = INITDIR(A\$)

Initializes a DECB directory listing. Call it before DIRPAGE. The pattern supports * for the rest of a filename field and ? for one character; use . for all files. A drive suffix is optional. No sequential file may be open. Returns 0 on success; disk failures produce a runtime disk error.

EXAMPLE

```
R = INITDIR("*.DAT:0")
```



x = DIRPAGE(0)

Gets the next directory page, containing up to sixteen records. Read the records with DIRLIST\$(0) through DIRLIST\$(15). Unused records on the last partial page contain sixteen zero bytes. Returns 0 for a retrieved page, including a final partial or empty page, and 4 when no listing is initialized or all 72 directory slots have been scanned.

EXAMPLE

```
R = DIRPAGE(0)
```



```
A$ = DIRLIST$(y)
```

Returns record y (0 through 15) from the most recent DIRPAGE. Bytes 1-8 are the filename, 9-11 the extension, byte 12 the attribute (\$20 for a regular file), and 13-16 the unsigned 32-bit size in most-significant-byte-first order. An unused record contains sixteen zero bytes. With fixed-length strings, A\$ must hold at least sixteen bytes.

EXAMPLE

```
A$ = DIRLIST$(3)
```



```
LOADM "filename.ext[:drive]"[,offset]
```

Loads a DECB machine-language file from floppy disk. The optional offset is added to every segment's load address. Files may contain more than 64 KB in total when composed of multiple segments; each segment address and length remains 16-bit.

EXAMPLE

```
LOADM "PROG.BIN:2",4231
```



```
SAVEM "filename.ext[:drive]",Start,End,Exec
```

Saves a memory range to a DECB floppy disk as a machine-language file. Start and End define the inclusive range; Exec is the run address.

EXAMPLE

```
SAVEM "LEVEL1.BIN:1",28000,29000,28000
```



Function Commands

ABS (n)

Returns the absolute value of n.

EXAMPLE

```
A = ABS(B)
```



ASC (string)

Returns the code (ASCII value) of the first character in string.

EXAMPLE

```
A = ASC(B$)
```



ATN (n)

Returns the arctangent of n, in radians.

EXAMPLE

```
A = ATN(B/3)
```



BUTTON (n)

Returns 1 if joystick button n is being pressed; returns 0 if it is not. n can be:

- 0 - Right joystick, Button 1 (old joystick)
- 1 - Right joystick, Button 2
- 2 - Left joystick, Button 1 (old joystick)
- 3 - Left joystick, Button 2

EXAMPLE

```
A = BUTTON(D)
```



CHR\$ (n)

Returns the character corresponding to character code n.

EXAMPLE

```
A$ = CHR$(65)
```



COCOHARDWARE(0)

Returns with a numeric value representing which CoCo you are using.

EXAMPLE

```
V = COCOHARDWARE(0)  
Where the bits of variable V will signify the CoCo  
Hardware as:  
Bit 0 is the Computer Type, 0 = CoCo 1/2, 1 = CoCo 3  
Bit 7 is the CPU type,      0 = 6809, 1 = 6309
```



COS (angle)

Returns the cosine of angle (angle is in radians).

EXAMPLE

```
A = COS(B)
```



EXP (n)

Returns the natural exponential of n (e^n).

EXAMPLE

```
A = EXP(B*1.15)
```



FIX (n)

Returns the truncated integer of n. Unlike INT, FIX does not return the next lower number for negative values; it truncates toward zero.

EXAMPLE

```
A = FIX(B - .2)
```



FujiNet functions

FujiNet functions use the **FN** prefix. They communicate through the CoCo Becker/DriveWire interface and are linked only when a program uses them. See the [FujiNet chapter](#) for setup, argument values, examples, limits, and error handling.

Network and JSON

Function	Result or operation
<code>FNINIT()</code>	Check for an accessible FujiNet; returns 0 on success.
<code>FNERROR()</code>	Return the most recent FujiNet library error.
<code>FNOPEN(channel, spec\$, access, translation)</code>	Open a network URL or device specification.
<code>FNCLOSE(channel)</code>	Close a network channel.
<code>FNBYTESWAITING(channel)</code>	Return the 16-bit number of bytes ready to read.
<code>FNCONNECTED(channel)</code>	Return the channel connected flag, 0 or 1.
<code>FNCHANNELERROR(channel)</code>	Return the channel protocol error byte.
<code>FNREAD\$(channel, count)</code>	Read up to 254 bytes into a BASIC string.
<code>FNWRITE(channel, data\$)</code>	Write a BASIC string.
<code>FNREADMEM(channel, address, count)</code>	Read directly into CoCo memory and return bytes transferred.
<code>FNWRITEMEM(channel, address, count)</code>	Write directly from CoCo memory.
<code>FNJSONPARSE(channel)</code>	Parse the current response body inside FujiNet.
<code>FNJSONQUERY\$(channel, path\$)</code>	Return the scalar value selected by a JSON path.

HTTP, filesystems, and authentication

Function	Result or operation
<code>FNHTTPMODE(channel, mode)</code>	Select HTTP body, header, or request-data mode.
<code>FNHTTPHEADER(channel, header\$)</code>	Add one HTTP request header.
<code>FNHTTPPOST(channel, data\$)</code>	Supply a POST request body.
<code>FNHTTPPUT(channel, data\$)</code>	Supply a PUT request body.
<code>FNHTTPDELETE(channel, url\$, translation)</code>	Open and execute an HTTP DELETE.
<code>FNUSERNAME(channel, user\$)</code>	Set the username for a channel.
<code>FNPASSWORD(channel, password\$)</code>	Set the password for a channel.
<code>FNDELETE(channel, spec\$)</code>	Delete a network file.
<code>FNRENAME(channel, spec\$)</code>	Rename a network file.
<code>FNLOCK(channel, spec\$)</code>	Make a network file read-only.
<code>FNUNLOCK(channel, spec\$)</code>	Make a network file writable.
<code>FNMKDIR(channel, spec\$)</code>	Create a network directory.
<code>FNRMDIR(channel, spec\$)</code>	Remove an empty network directory.
<code>FNCHDIR(channel, spec\$)</code>	Change the N: prefix/current directory.
<code>FNDIROPEN(channel, spec\$, format)</code>	Open a network-directory stream.

Wi-Fi, hosts, devices, and mounting

Function	Result or operation
<code>FNWIFISTATUS()</code>	Return Wi-Fi status; 3 means connected.
<code>FNWIFIENABLED()</code>	Return 1 when Wi-Fi is enabled.
<code>FNWIFISCAN()</code>	Scan and return the number of visible networks.
<code>FNWIFISSID\$(index)</code>	Return a scanned network name.
<code>FNWIFIRSSI(index)</code>	Return a scanned network's signed signal strength.
<code>FNSETWIFI(ssid\$, password\$)</code>	Save and connect to a wireless network.
<code>FNADAPTER\$(field)</code>	Return one adapter-information field.
<code>FNHOST\$(slot)</code>	Return a host-slot name.
<code>FNSETHOST(slot, name\$)</code>	Change a host-slot name.
<code>FNHOSTPREFIX\$(slot)</code>	Return a host-slot path prefix.
<code>FNSETHOSTPREFIX(slot, prefix\$)</code>	Change a host-slot path prefix.
<code>FNMOUNTHOST(slot)</code>	Mount a host slot.
<code>FNUNMOUNTHOST(slot)</code>	Unmount a host slot.
<code>FNDEVICEPATH\$(device)</code>	Return a device-slot path.
<code>FNDEVICEHOST(device)</code>	Return a device's host-slot number.
<code>FNDEVICEMODE(device)</code>	Return a device's stored mode byte.
<code>FNSETDEVICE(device, host, mode, path\$)</code>	Configure a disk-device slot.

Function	Result or operation
<code>FNMountImage(device, mode)</code>	Mount an image in a device slot.
<code>FNUnmountImage(device)</code>	Unmount an image.
<code>FNMountAll()</code>	Attempt to mount all configured devices.

Fuji host directory, clock, and data services

Function	Result or operation
<code>FNOPENDIR(host, path\$, filter\$)</code>	Open a directory on a configured Fuji host.
<code>FNREADDIR\$(maxlen, details)</code>	Return the next directory entry.
<code>FNDIRPOS()</code>	Return the current directory position.
<code>FNSETDIRPOS(position)</code>	Seek to a directory position.
<code>FNCLOSEDIR()</code>	Close the global Fuji directory handle.
<code>FNTIME\$()</code>	Return FujiNet-local time as <code>YYYY-MM-DD HH:MM:SS</code> .
<code>FNAPPKEYSET(creator, app)</code>	Select an AppKey creator ID and application ID.
<code>FNAPPKEYREAD\$(key)</code>	Read up to 64 bytes from an AppKey.
<code>FNAPPKEYWRITE(key, data\$)</code>	Write up to 64 bytes to an AppKey.
<code>FNAPPKEYCLOSE()</code>	Close the selected AppKey.
<code>FNHASH\$(algorithm, data\$)</code>	Return a lowercase hexadecimal digest.
<code>FNHASHCLEAR()</code>	Clear accumulated hash input.
<code>FNHASHADD(data\$)</code>	Append data to accumulated hash input.
<code>FNHASHCALC\$(algorithm, discard)</code>	Hash accumulated input.
<code>FNBASE64ENCODE\$(data\$)</code>	Base64 encode up to 189 bytes.
<code>FNBASE64DECODE\$(data\$)</code>	Decode up to 252 bytes of valid Base64.

HEX\$ (n)

Returns a string containing the hexadecimal value of n.

EXAMPLE

```
PRINT HEX$(A); "="; A
```



INKEY\$

Checks the keyboard and returns the key being pressed or, if no key is being pressed, returns a null string ("").

EXAMPLE

```
A$ = INKEY$
```



INSTR (p, s, t)

Searches a string. Returns the position of target string t, within search string s, starting at position p.

- p - Start position of search
- s - String being searched
- t - Target string

EXAMPLE

```
A = INSTR(1, M5$, "BEETS")
```



INT (n)

Converts n to the largest integer that is less than or equal to n.

EXAMPLE

```
A = INT(B + .5)
```



JOYSTK (i)

Returns the horizontal or vertical coordinate (i) of the left or right joystick.

- 0 - Horizontal, right joystick
- 1 - Vertical, right joystick
- 2 - Horizontal, left joystick
- 3 - Vertical, left joystick

EXAMPLE

```
A = JOYSTK(B)
```



LEFT\$ (string, length)

Returns the left portion of a string. length specifies the number of characters returned.

EXAMPLE

```
A$ = LEFT$(B$, 3)
```



LEN (string)

Returns the length of string.

EXAMPLE

```
A = LEN(B$)
```



LOG (n)

Returns the natural logarithm of n.

EXAMPLE

```
A = LOG(B / 2)
```



LPEEK (memory location)

Returns the contents of a virtual memory location (0-2097151 decimal or 0-&H1FFFFFF hex)

EXAMPLE

```
A = LPEEK(&H60000)
```



LTRIM\$ (string)

Removes both leading spaces from the string.

EXAMPLE

```
A$ = LTRIM$(B$)
```



MID\$ (s, p, l)

Returns a substring of string s.

- s - Source string
- p - Starting position of substring
- l - Length of substring

EXAMPLE

```
A$ = MID$(B$, 2, 2)
```



PEEK (memory location)

Returns the contents of a memory location (0-65535 decimal or 0-&HFFFF hexadecimal).

EXAMPLE

```
A = PEEK(3020)
```



POINT (x,y)

Returns the colour value of the pixel selected from the current GMODE screen.

- x,y - Screen location of the pixel value requested

EXAMPLE

```
A = POINT(10,12)
```



POS (dev)

Returns the current print position. dev (print device number): -2 - Printer

- 0 - Screen

EXAMPLE

```
A = POS(0)
```



RIGHT\$ (string, length)

Returns the right portion of a string. length specifies the number of characters returned.

EXAMPLE

```
A$ = RIGHT$(B$, 4)
```



RND (n)

Generates a "random" number between 1 and n. If n = 0 then it generates a random number from 0 to <1

EXAMPLE

```
A = RND(0)
```



RTRIM\$ (string)

Removes trailing spaces from the string.

EXAMPLE

```
A$ = RTRIM$(B$)
```



SGN (n)

Returns the sign of n: -1 - Negative

- 0 - Zero
- 1 - Positive

EXAMPLE

```
A = SGN(A + .1)
```



SIN (angle)

Returns the sine of angle (angle is in radians).

EXAMPLE

```
A = SIN(B/3.14159)
```



STRING\$ (l, c)

Returns a string of a repeated character.

- l - Length of string
- c - Character used (can be a code or a string)

EXAMPLE

```
A$ = STRING$(22, "A")
```



STR\$ (n)

Converts n to a string.

EXAMPLE

```
A$ = STR$(1.234)
```



SQR (n)

Returns the square root of n.

EXAMPLE

```
A = SQR(B/2)
```



TAN (angle)

Returns the tangent of angle (angle is in radians).

EXAMPLE

```
A = TAN(B)
```



TIMER

Returns the contents of the timer (0–65535).

EXAMPLE

```
A = TIMER / 18
```



TRIM\$ (string)

Removes both leading and trailing spaces from the string.

EXAMPLE

```
A$ = TRIM$(B$)
```



VAL (string)

Converts a string to a number.

EXAMPLE

```
A = VAL("1.23")
```



VARPTR (variable)

Returns a pointer to where a variable is located in memory.

EXAMPLE

```
A = VARPTR(B)
```

You cannot change where a variable is in memory, you can only use this to find out where it exists.



Constants

You can use Constants in your program that will act like place holders for actual values that will be substituted at compile time. They are different than variables as they don't actually take any RAM in your program.

```
Examples:  
Const PI = 3.141593  
Const PI2 = PI * 2
```



Constants can also be quoted strings:

```
Const circumferenceText = "THE CIRCUMFERENCE OF THE CIRCLE IS"  
Const areaText = "THE AREA OF THE CIRCLE IS"
```



Math / Logical Operators

Operator	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division (Integers will be rounded)
^	Exponent
MOD	Remainder of division
\	Integer division (no rounding)
AND	Bitwise AND
OR	Bitwise OR
XOR	Bitwise Exclusive OR
NOT	Bitwise Compliment (invert each bit)

FujiNet

BasTo6809 programs can use FujiNet networking and service features through the CoCo Becker/DriveWire interface. The compiler automatically links only the FujiNet assembly modules referenced by the program, so BASIC programs that do not use an `FN` function pay no FujiNet RAM cost.

FujiNet-PC, MAME with Becker support, and a compatible physical FujiNet are all valid targets. Network channels are numbered 0 through 7. Unless an entry says otherwise, a numeric function returns 0 on success and a nonzero error on failure. `FNERROR()` returns the most recent library error.

Starting and testing

Always check FujiNet before opening a network resource:

```
E = FNINIT()  
If E <> 0 Then Print "FUJINET NOT AVAILABLE"; E: End
```

`FNINIT()` performs an actual FujiNet status transaction. It is not a test of whether a URL or remote server is available. Test the return value from `FNOPEN`, then use `FNCHANNELERROR`, `FNCONNECTED`, and `FNBYTESWAITING` to diagnose the selected channel.

To make F11 compile and launch the program in MAME with the Becker interface, place this directive near the beginning of the BASIC source:

```
' F11EMULATOR FUJINET
```

Start FujiNet-PC first. The supplied makefile verifies that TCP port 65504 is listening, starts MAME with its debugger, and uses Becker registers `$FF41` (status) and `$FF42` (data). From a terminal, `make FujiNetDebug` starts the same FujiNet-aware debugger configuration without compiling a BASIC program.

Network channels

Function	Purpose
<code>FNOPEN(channel, spec\$, access, translation)</code>	Open a URL or device specification.
<code>FNCLOSE(channel)</code>	Close one channel.
<code>FNBYTESWAITING(channel)</code>	Return the 16-bit number of readable bytes.
<code>FNCONNECTED(channel)</code>	Return the connected flag, 0 or 1.
<code>FNCHANNELERROR(channel)</code>	Return the protocol error byte.
<code>FNREAD\$(channel, count)</code>	Read up to 254 bytes into a BASIC string.
<code>FNWRITE(channel, data\$)</code>	Write a BASIC string.
<code>FNREADMEM(channel, address, count)</code>	Read binary data directly into CoCo memory.
<code>FNWRITEMEM(channel, address, count)</code>	Write binary data directly from CoCo memory.

Common `FNOPEN` access values are:

Access	Meaning
4	GET or read
5	HTTP DELETE
6	Directory read
8	PUT or write
12	GET with response headers
13	POST

Translation values are 0 for none, 1 for CR, 2 for LF, 3 for CR/LF, and 4 for PET translation. URL schemes supported by the installed FujiNet firmware can be used in `spec$`, normally with the `N:` device prefix.

Do not request more than the current `FNBYTESWAITING` result from `FNREAD$`, and do not request more than 254 bytes in one call. Loop for a longer response. For binary transfers, `FNREADMEM` and `FNWRITEMEM` accept a 16-bit count and reject zero-length or wrapping address ranges. The selected memory must not overwrite the program, stack, runtime, or hardware registers.

Error handling

An open channel can fail even when `FNINIT()` succeeds. A robust program checks each operation and closes any channel that it opened:

```
E = FNOPEN(0, URL$, 4, 0)
If E <> 0 Then Print "OPEN ERROR"; E: GoTo 900

W = FNBYTESWAITING(0)
If FNCHANNELERROR(0) <> 0 Then Print "CHANNEL ERROR"; FNCHANNELERROR(0)
If W > 254 Then W = 254
If W > 0 Then Reply$ = FNREAD$(0, W)

E = FNCLOSE(0)
900 End
```

`FNCONNECTED(channel)` reports the protocol's connection flag. A value of 0 can be normal after an HTTP response completes, so interpret it together with the channel error and bytes-waiting values.

JSON

Function	Purpose
<code>FNJSONPARSE(channel)</code>	Parse the current response body inside FujiNet.
<code>FNJSONQUERY\$(channel, path\$)</code>	Return one scalar selected by a JSON path.

JSON parsing occurs in FujiNet and does not require a JSON document buffer in CoCo RAM. Open the URL for reading, call `FNJSONPARSE`, then query individual paths. Array indexes appear as path components, such as `"/daily/temperature_2m_max/1"`.

Public weather example

This program reads live public data and does not need a local Python server:

```
'compileoptions -s16
'F11EMULATOR FUJINET

Dim URL$ As String * 254
Dim Temperature$ As String * 254
Dim Humidity$ As String * 254
Dim Tomorrow$ As String * 254
Dim High$ As String * 254

URL$ = "N:HTTPS://api.open-meteo.com/v1/forecast?
latitude=43.6532&longitude=-79.3832&current=temperature_2m,relative_humidity_2m&daily=te
mperature_2m_max&forecast_days=2"

E = FNINIT()
If E <> 0 Then Print "FUJINET ERROR"; E: End

E = FNOPEN(0, URL$, 4, 0)
If E <> 0 Then Print "OPEN ERROR"; E: End

E = FNJSONPARSE(0)
If E <> 0 Then Print "PARSE ERROR"; E: C = FNCLOSE(0): End

Temperature$ = FNJSONQUERY$(0, "/current/temperature_2m")
Humidity$ = FNJSONQUERY$(0, "/current/relative_humidity_2m")
Tomorrow$ = FNJSONQUERY$(0, "/daily/time/1")
High$ = FNJSONQUERY$(0, "/daily/temperature_2m_max/1")

If FNERROR() <> 0 Then
    Print "QUERY ERROR"; FNERROR()
Else
    Print "TORONTO TEMP: "; Temperature$; " C"
    Print "HUMIDITY: "; Humidity$; "%"
    Print Tomorrow$; " HIGH: "; High$; " C"
End If

C = FNCLOSE(0)
End
```

HTTP

Function	Purpose
<code>FNHTTPMODE(channel, mode)</code>	Select a raw HTTP channel mode.
<code>FNHTTPHEADER(channel, header\$)</code>	Add one request header.
<code>FNHTTPPOST(channel, data\$)</code>	Supply the body for access mode 13.
<code>FNHTTPPUT(channel, data\$)</code>	Supply the body for access mode 8.
<code>FNHTTPDELETE(channel, url\$, translation)</code>	Open and execute a DELETE request.

HTTP modes are 0 body, 1 select response headers, 2 read collected headers, 3 set request headers, and 4 supply POST/PUT data. Header strings use the form `"Name: value"`.

```
E = FOPEN(0, "N:HTTPS://example.com/items", 13, 0)
If E = 0 Then E = FNHTTPHEADER(0, "Content-Type: application/json")
If E = 0 Then E = FNHTTPPOST(0, "{""name"":""CoCo""}")
C = FCLOSE(0)
```

Current FujiNet firmware treats documented header-capable PUT access mode 14 as POST. Use access mode 8 for PUT until that firmware behavior is corrected.

Network filesystems and authentication

Function	Purpose
<code>FNUSERNAME(channel, user\$)</code>	Set a username before opening the channel.
<code>FNPASSWORD(channel, password\$)</code>	Set a password before opening the channel.
<code>FNDELETE(channel, spec\$)</code>	Delete a file.
<code>FNRENAME(channel, spec\$)</code>	Rename using <code>old-path,new-name</code> in one specification.
<code>FNLOCK(channel, spec\$)</code>	Make a file read-only.
<code>FNUNLOCK(channel, spec\$)</code>	Make a file writable.
<code>FNMKDIR(channel, spec\$)</code>	Create a directory.
<code>FNRMDIR(channel, spec\$)</code>	Remove an empty directory.
<code>FNCHDIR(channel, spec\$)</code>	Change the N: prefix/current directory.
<code>FNDIROPEN(channel, spec\$, format)</code>	Open a network-directory stream.

Useful directory formats are 0 for shortened text, `&H80` for long CoCo text, and `&H83` for raw names, where directories end in `/`. Read directory data with `FNBYTESWAITING` and `FNREAD$`.

Filesystem operations depend on the URL backend. TNFS implements the full set; other protocols may implement only a subset. Lock and unlock are reliably meaningful only on TNFS in current firmware.

Wi-Fi and adapter information

Function	Purpose
<code>FNWIFISTATUS()</code>	Return connection status; 3 means connected.
<code>FNWIFIENABLED()</code>	Return 1 when Wi-Fi is enabled.
<code>FNWIFISCAN()</code>	Scan and return the visible SSID count.
<code>FNWIFISSID\$(index)</code>	Return one scanned SSID.
<code>FNWIFIRSSI(index)</code>	Return its signed signal strength.
<code>FNSETWIFI(ssid\$, password\$)</code>	Save and connect to an SSID.
<code>FNADAPTER\$(field)</code>	Return one adapter-information field.

`FNADAPTER$` fields are 0 SSID, 1 hostname, 2 local IP, 3 gateway, 4 netmask, 5 DNS, 6 MAC, 7 BSSID, and 8 firmware version.

FujiNet-PC has a dummy Wi-Fi manager. Its scans return test networks, and setting an SSID changes `fnconfig.ini`; it does not configure the Mac's Wi-Fi.

Host slots, device slots, and mounting

Function	Purpose
<code>FNHOST\$(slot) / FNSETHOST(slot, name\$)</code>	Read or change a host slot.
<code>FNHOSTPREFIX\$(slot) / FNSETHOSTPREFIX(slot, prefix\$)</code>	Read or change its path prefix.
<code>FNMOUNTHOST(slot) / FNUNMOUNTHOST(slot)</code>	Mount or unmount a host.
<code>FNDEVICEPATH\$(device)</code>	Return the full device-slot path.
<code>FNDEVICEHOST(device)</code>	Return its host-slot number.
<code>FNDEVICEMODE(device)</code>	Return its stored mode byte.
<code>FNSETDEVICE(device, host, mode, path\$)</code>	Configure a disk-device slot.
<code>FNMOUNTIMAGE(device, mode) / FNUNMOUNTIMAGE(device)</code>	Mount or unmount an image.
<code>FNMOUNTALL()</code>	Attempt to mount all configured devices.

The CoCo firmware has eight host slots numbered 0 through 7 and four disk device slots numbered 0 through 3. Device mode 1 is read-only and 2 is writable. Bit `&H40` in a returned device mode indicates mounted state. `FNMOUNTALL()` can partially succeed, so check `FNERROR()` and the individual device slots if recovery matters.

Fuji host directory browser

This directory API browses a configured host slot and is separate from the network-channel `FNDIROPEN` function.

Function	Purpose
<code>FNOPENDIR(host, path\$, filter\$)</code>	Open a directory on a host slot.
<code>FNREADDIR\$(maxlen, details)</code>	Return the next entry.
<code>FNDIRPOS()</code>	Return the cursor position.
<code>FNSETDIRPOS(position)</code>	Seek the cursor.
<code>FNCLOSEDIR()</code>	Close the global directory handle.

Clock

`FNTIME$()` returns FujiNet-local time as `YYYY-MM-DD HH:MM:SS`. FujiNet-PC uses the Mac's clock and timezone. Timezone setters are not exposed because the current CoCo DriveWire firmware path swaps their multibyte parameters.

AppKey storage

Function	Purpose
<code>FNAPPKEYSET(creator, app)</code>	Select nonzero 16-bit creator and application IDs.
<code>FNAPPKEYREAD\$(key)</code>	Read up to 64 binary-safe bytes.
<code>FNAPPKEYWRITE(key, data\$)</code>	Write up to 64 bytes.
<code>FNAPPKEYCLOSE()</code>	Close the selected AppKey.

AppKeys are persistent application storage, not secure secret storage. FujiNet stores them as unencrypted files on its SD filesystem.

Hashing and Base64

Function	Purpose
<code>FNHASH\$(algorithm, data\$)</code>	Return a lowercase hexadecimal digest.
<code>FNHASHCLEAR()</code>	Clear accumulated hash input.
<code>FNHASHADD(data\$)</code>	Append binary-safe input.
<code>FNHASHCALC\$(algorithm, discard)</code>	Hash accumulated input, optionally discarding it.
<code>FNBASE64ENCODE\$(data\$)</code>	Base64 encode up to 189 bytes.
<code>FNBASE64DECODE\$(data\$)</code>	Decode up to 252 bytes of valid Base64.

Hash algorithms are 0 MD5, 1 SHA-1, 2 SHA-256, and 3 SHA-512. All hexadecimal digests fit in a 254-byte BASIC string. `discard` is 0 to keep accumulated hash input or 1 to discard it after calculating the digest. Base64 encode input is limited to 189 bytes so the expanded result fits in a 254-byte BASIC string.

Operational notes

- Setting Wi-Fi, hosts, prefixes, device slots, and AppKeys changes persistent FujiNet configuration or storage. Check arguments before running test code.
- Close channels and directory handles even after a failed operation whenever they may have been opened successfully.
- FujiNet capabilities can vary by firmware build and URL backend. A successful low-level command does not guarantee that every backend implements the requested filesystem operation.
- String-returning functions are limited by the BASIC string size selected by `-s`; these functions support a maximum of 254 data bytes.

Firmware references

- [FujiNet network API](#)

- [Fuji device API](#)
- [FujiNet CoCo implementation](#)

CoCoMP3 Commands

You can easily and directly control the CoCoMP3 hardware from your BASIC program using the following commands:

COCOMP3_AUDIO_MODE(x)

Sets where CoCoMP3 audio is routed. x=0 leave unchanged x=1 (default) route audio to the CoCo cassette interface so it plays through the TV speaker.

EXAMPLE

```
z=COCOMP3_AUDIO_MODE(1)
```



COCOMP3_COMBINATION_PLAY_SETTING(x,y)

Sets a "combination play" range of tracks from x to y.

EXAMPLE

```
z=COCOMP3_COMBINATION_PLAY_SETTING(10,25)
```



COCOMP3_CYCLE_MODE_SETTING(x)

Sets playback mode. x=0 full cycle 1 repeat track 2 stop after track 3 random all 4 repeat folder 5 random in folder 6 order in folder then stop 7 order all then stop.

EXAMPLE

```
z=COCOMP3_CYCLE_MODE_SETTING(4)
```



COCOMP3_END_COMBINATION_PLAY(0)

When a combination play range finishes, stop playback.

EXAMPLE

```
z=COCOMP3_END_COMBINATION_PLAY(0)
```



COCOMP3_END_PLAYING(0)

Ends the current track and skips to the next track (like Next).

EXAMPLE

```
z=COCOMP3_END_PLAYING(0)
```



COCOMP3_GET_CURRENT_TRACK(0)

Returns the current track number assigned by the CoCoMP3 (usable with PLAY_TRACK_NUMBER or SELECT_BUT_NO_PLAY).

EXAMPLE

```
z=COCOMP3_GET_CURRENT_TRACK(0)
```



COCOMP3_GET_DRIVE_STATUS(0)

Returns microSD status:

- 0 = not installed
- 2 = installed

EXAMPLE

```
z=COCOMP3_GET_DRIVE_STATUS(0)
```



COCOMP3_GET_FOLDER_DIR_TRACK(0)

Returns the first track number in the current folder.

EXAMPLE

```
z=COCOMP3_GET_FOLDER_DIR_TRACK(0)
```



COCOMP3_GET_NUMBER_OF_TRACKS(0)

Returns the total number of tracks on the microSD.

EXAMPLE

```
z=COCOMP3_GET_NUMBER_OF_TRACKS(0)
```



COCOMP3_GET_PLAY_STATUS(0)

Returns play status:

- 0 = stopped
- 1 = playing

EXAMPLE

```
z=COCOMP3_GET_PLAY_STATUS(0)
```



COCOMP3_GET_TRACKS_IN_FOLDER(0)

Returns how many tracks are in the current folder.

EXAMPLE

```
z=COCOMP3_GET_TRACKS_IN_FOLDER(0)
```



COCOMP3_NEXT(0)

Skips to (and plays) the next track.

EXAMPLE

```
z=COCOMP3_NEXT(0)
```



COCOMP3_PAUSE(0)

Pauses playback of the current track.

EXAMPLE

```
z=COCOMP3_PAUSE(0)
```



COCOMP3_PLAY(0)

Plays the current track (or the first track after power-on).

EXAMPLE

```
z=COCOMP3_PLAY(0)
```



COCOMP3_PLAY_NEXT_FOLDER(0)

Advances to the next folder and starts playing there (behavior depends on Cycle Mode).

EXAMPLE

```
z=COCOMP3_PLAY_NEXT_FOLDER(0)
```



COCOMP3_PLAY_PREVIOUS_FOLDER(0)

Moves to the previous folder and starts playing there (behavior depends on Cycle Mode).

EXAMPLE

```
z=COCOMP3_PLAY_PREVIOUS_FOLDER(0)
```



COCOMP3_PLAY_TRACK(T\$)

Selects and plays a specific file path. The command converts ".MP3/.WAV" to "MP3/WAV" as required by the hardware.

EXAMPLE

```
z=COCOMP3_PLAY_TRACK("/FOLDER02/00001.MP3")
```



COCOMP3_PLAY_TRACK_NUMBER(x)

Plays track number x (track numbers follow copy order to the microSD; use GET_CURRENT_TRACK to discover a track's number).

EXAMPLE

```
z=COCOMP3_PLAY_TRACK_NUMBER(12)
```



COCOMP3_PREVIOUS(0)

Skips to (and plays) the previous track.

EXAMPLE

```
z=COCOMP3_PREVIOUS(0)
```



COCOMP3_RAW(R\$)

Sends a raw command packet to the CoCoMP3 (for advanced control; see DY-SV5W_Datasheet.pdf for codes).

EXAMPLE

```
z=COCOMP3_RAW(CHR$(&HAA)+CHR$(1)+CHR$(0)+CHR$(&HAB))
```



COCOMP3_SELECT_BUT_NO_PLAY(x)

Stops the current track, queues track x, but does not start playback until you call PLAY.

EXAMPLE

```
z=COCOMP3_SELECT_BUT_NO_PLAY(12)
```



COCOMP3_SET_CYCLE_TIMES(x)

Sets how many times to repeat the current track/folder before stopping; manually selecting another track overrides this.

EXAMPLE

```
z=COCOMP3_SET_CYCLE_TIMES(3)
```



COCOMP3_SET_EQ(x)

Sets the internal EQ preset: 0 Normal 1 Pop 2 Rock 3 Jazz 4 Classical

EXAMPLE

```
z=COCOMP3_SET_EQ(2)
```



COCOMP3_SET_PATH_INTERLUDE(P\$)

Interrupts the current track, plays tracks from folder path P\$, then resumes the interrupted track afterward (behaviour may depend on Cycle Mode).

EXAMPLE

```
z=COCOMP3_SET_PATH_INTERLUDE("/FOLDER02/*MP3")
```



COCOMP3_SET_TRACK_INTERLUDE(x)

Interrupts the current track, plays track x, then resumes the interrupted track afterward.

EXAMPLE

```
z=COCOMP3_SET_TRACK_INTERLUDE(20)
```



COCOMP3_SET_VOL(x)

Sets volume level 0–30 (default on power-up is 30).

EXAMPLE

```
z=COCOMP3_SET_VOL(18)
```



COCOMP3_STOP(0)

Stops playback of the current track.

EXAMPLE

```
z=COCOMP3_STOP(0)
```



COCOMP3_TEST(0)

Checks if CoCoMP3 is ready. Returns: 0 OK -1 not powered/connected -2 no microSD -3 no tracks on microSD.

EXAMPLE

```
z=COCOMP3_TEST(0)
```



COCOMP3_VOL_DOWN(0)

Decreases volume by 1 step.

EXAMPLE

```
z=COCOMP3_VOL_DOWN(0)
```



COCOMP3_VOL_FADE(x)

Fades volume to 0 over x milliseconds, then stops playback (software-driven; the CoCo is halted during the fade).

EXAMPLE

```
z=COCOMP3_VOL_FADE(1500)
```



COCOMP3_VOL_MAX(0)

Sets volume to maximum (30).

EXAMPLE

```
z=COCOMP3_VOL_MAX(0)
```



COCOMP3_VOL_UP(0)

Increases volume by 1 step.

EXAMPLE

```
z=COCOMP3_VOL_UP(0)
```



CoCo SDC Specific Commands

These commands access the CoCoSDC FAT32 filesystem directly from BASIC programs. File numbers and CoCoSDC slots are 0 or 1.

```
SDC_CHAIN "FILENAME.BIN",#
```

Loads a DECB-format machine-language program directly from the CoCoSDC FAT filesystem, replaces the current program, and executes its LOADM postamble address. Include the extension. SDC_CHAIN does not return.

The target must be the normal `.BIN` produced by assembling the compiler's output. Do not process a program that will be loaded by `SDC_CHAIN` with `cc1sl`; `SDC_CHAIN` already contains the loader needed to load a large program across the 64K address space. A compiler-generated first program containing `SDC_CHAIN` should be processed with `cc1sl` when Disk BASIC loads it, even if its main code is small, because the resident loader occupies part of Disk BASIC's loading workspace. See [Preparing programs for CHAIN and SDC_CHAIN](#) for the complete build procedure.

EXAMPLE

```
SDC_CHAIN "PART2.BIN",0
```



```
SDC_LOADM "FILENAME.BIN",#[,Offset]
```

Loads a machine-language binary from the CoCoSDC. Offset is optionally added to the original LOADM addresses.

EXAMPLE

```
SDC_LOADM "GAME.BIN",0,256
```



```
x = _SDC_FILEEXISTS(string,#)
```

Returns 255 (-1) when the file exists in the CoCoSDC FAT filesystem, or 0 when it does not.

EXAMPLE

```
A = _SDC_FILEEXISTS(N$,0)
```



SDC_SAVEM "FILENAME.BIN",#,Start,End,Exec

Saves a memory range directly to the CoCoSDC as a machine-language file. Start and End define the inclusive range; Exec is the run address.

EXAMPLE

```
SDC_SAVEM "LEVEL1.BIN",1,28000,29000,28000
```



SDC_BIGLOADM "FILENAME.BIN",#

Loads CoCo 3 memory blocks quickly from a specially prepared BIGLOADM file. This is useful for large assets such as screen backgrounds.

EXAMPLE

```
SDC_BIGLOADM "TITLESER.BIN",0
```



SDC_OPEN "FILENAME.EXT","X",#

Opens a CoCoSDC file for direct access. Use R for reading or W for writing.

EXAMPLE

```
SDC_OPEN "DATA.DAT","R",0
```



SDC_CLOSE(#)

Closes open CoCoSDC file 0 or 1.

EXAMPLE

```
SDC_CLOSE(0)
```



```
SDC_PUTBYTE0 x / SDC_PUTBYTE1 x
```

Writes one byte to open output file 0 or 1. The corresponding file position advances automatically.

EXAMPLE

```
SDC_PUTBYTE0 A
```



```
x = SDC_GETBYTE(#)
```

Reads one byte from file 0 or 1. The file position advances automatically.

EXAMPLE

```
B = SDC_GETBYTE(0)
```



```
x = SDC_LOF(#)
```

Returns the open file's length as an unsigned 32-bit value (`_UNDEFINED LONG`) without changing its current position.

EXAMPLE

```
FileSize = SDC_LOF(0)
```



```
SDC_SETPOS0(x) / SDC_SETPOS1(x)
```

Sets the corresponding open file's zero-based position to unsigned 32-bit byte offset `x`. On a file opened for reading, this changes only the read position and writes nothing.

EXAMPLE

```
SDC_SETPOS0(299)
```



```
A$ = SDC_FILEINFO$(#)
```

Returns a 32-byte information record. Bytes 1-8 are the name, 9-11 the extension, byte 12 the attributes, and 29-32 the size in least-significant-byte-first order.

EXAMPLE

```
A$ = SDC_FILEINFO$(0)
```

```
x = SDC_DELETE(A$)
```

Deletes a file or empty directory at full path A\$. Returns 0 for success, 1 if busy too long, 3 for an invalid path, 4 for a hardware error, 5 when not found, or 6 when a directory is not empty.

EXAMPLE

```
R = SDC_DELETE("/GAMES/OLD.BIN")
```

```
x = SDC_MKDIR(A$)
```

Creates the directory named by full path A\$. Returns 0 for success, 1 if busy too long, 3 for an invalid path, 4 for a hardware error, 5 when the parent is not found, or 6 when the name is already in use.

EXAMPLE

```
R = SDC_MKDIR("/GAMES/NEW")
```

```
x = SDC_SETDIR(A$)
```

Changes the current CoCoSDC directory to full path A\$. Returns 0 for success, 1 if busy too long, 3 for an invalid path, 4 for a hardware error, or 5 when the directory is not found.

EXAMPLE

```
R = SDC_SETDIR("/GAMES")
```

```
A$ = SDC_GETCURDIR$(#)
```

Returns current-directory information for slot 0 or 1. Bytes 1-8 are the filename, 9-11 the extension, and 12-31 are private or reserved.

EXAMPLE

```
A$ = SDC_GETCURDIR$(0)
```

```
x = SDC_INITDIR(A$)
```

Initializes a directory listing using a path and wildcard filter. Call it before SDC_DIRPAGE and SDC_DIRLIST\$. Returns 0 for success, 1 if busy too long, 3 for an invalid path, 4 for a hardware error, or 5 when the directory is not found.

EXAMPLE

```
R = SDC_INITDIR("MYDIR/*.TXT")
```

```
x = SDC_DIRPAGE(0)
```

Retrieves the next 256-byte directory page: sixteen 16-byte records. Continue until a page contains unused zero records. Returns 0 for success, 1 if busy too long, or 4 when not initialized or at end of listing.

EXAMPLE

```
R = SDC_DIRPAGE(0)
```

```
A$ = SDC_DIRLIST$(y)
```

Returns entry y (0 through 15) from the latest page. Bytes 1-8 are the name, 9-11 the extension, byte 12 the attributes, and 13-16 the size in most-significant-byte-first order.

EXAMPLE

```
A$ = SDC_DIRLIST$(3)
```

CoCo SDC Movie Playback Commands

`SDC_PLAYMOVIE` streams an NTM movie from a CoCoSDC and selects the graphics mode recorded in the movie header.

```
SDC_PLAYMOVIE "filename",#
```

- `filename` is the NTM file on the CoCoSDC filesystem.
- `#` is CoCoSDC file slot 0 or 1.

EXAMPLE

```
SDC_PLAYMOVIE "MYMOVIE.NTM",0
```



During playback:

- Press `1` through `9` to jump to 10 through 90 percent of the movie.
- Press Left Arrow to move back 60 seconds.
- Press Right Arrow to move forward 120 seconds.
- Press Space to pause or resume.
- Press BREAK to stop playback and return to the BASIC program.

On a CoCo 1/2, the movie player double-buffers two \$1800-byte frames. The compiler therefore reserves \$3000 bytes of graphics memory whenever a CoCo 1/2 movie mode (GMODE 8, 15, or 16) is used with `SDC_PLAYMOVIE`, even if the BASIC source selects page 0. The buffers follow `GRAPHICS_PAGE_START`, so `-notext`, `-rxxxx`, and resident CHAIN loaders are included in the placement calculation. Playback restores the program's selected graphics page before returning.

This movie-only reservation does not propagate to a CHAIN target. A target that does not play movies and only uses `GMODE 16,0` reserves one \$1800-byte page and can reuse the former second movie buffer. Reserved `-rxxxx` shared RAM remains available for parameters when all programs use compatible layout options.

Creating NTM files

Use `MakeNTM` to convert numbered PNG frames or a source movie. Direct movie input requires `ffmpeg` and `ffprobe` in the command path unless their paths are provided explicitly. The `-g` option selects the GMODE stored in the NTM file.

```
./MakeNTM -i=input.mp4 -o=movie.ntm -g16 -t2 --start=00:05:00 --length=00:01:00  
./MakeNTM -i=input.mov -o=movie.ntm -g136 -f12 -a2 -p1 -c1 -r=256x144
```



See `MakeNTM_User_Guide.txt` for the complete format, mode, audio, cropping, scaling, and command-line option reference.

CoCo SDC Audio Playback Commands

These commands allow you to playback audio samples/music files directly off the CoCo SDC at a very high bitrate. For mono at 44.75 kHz and stereo at 22.375 kHz

```
SDC_PLAY "filename",#
```

Plays an audio file stored on the SDC, outputting sound directly through the CoCo. # is the file number (0 or 1).

EXAMPLE

```
SDC_PLAY "MYAUDIO.RAW",0
```



```
SDC_PLAYORCL "filename",#
```

Plays an audio file stored on the SDC, outputting sound through the left channel of the Orchestra 90/CoCo Flash # is the file number (0 or 1).

EXAMPLE

```
SDC_PLAYORCL "MYAUDIO.RAW",0
```



```
SDC_PLAYORCR "filename",#
```

Plays an audio file stored on the SDC, outputting sound directly through the CoCo. # is the file number (0 or 1).

EXAMPLE

```
SDC_PLAYORCR "MYAUDIO.RAW",0
```



```
SDC_PLAYORCS "filename",#
```

Plays an audio file stored on the SDC, outputting sound directly through the CoCo. # is the file number (0 or 1).

EXAMPLE

SDC_PLAYORCS "MYAUDIO.RAW",0



To stop a playing music file or sample you can press the BREAK key.

In order for you to get your audio sample in the correct format to be played back you'll need to prepare your audio samples and put them on the SD card. The format for the raw audio file that will be played is mono 8 bits unsigned. To convert any sound file or even the audio from a video file to the correct format used with the SDCPLAY command use FFMPEG and the following command:

```
ffmpeg -i source_audio.mp3 -acodec pcm_u8 -f u8 -ac 1 -ar 44750 -af  
aresample=44750:filter_size=256:cutoff=1.0 MYAUDIO.RAW
```

You can also use an audio tool like Audacity to Export an audio sample to an uncompressed Mono 44750 Hz, RAW (header-less) unsigned 8-bit PCM sample.

If you want to stream 8 bit stereo sound from your CoCo to the COCOFLASH/Orchestra90 use the command:

SDC_PLAYORCS where the sample MYSAMPLE.RAW is stored on the SD card in your SDC Controller. It can be created with the FFMPEG command below:

```
ffmpeg -i source_audio.mp3 -acodec pcm_u8 -f u8 -ac 2 -ar 22375 -af  
aresample=22375:filter_size=256:cutoff=1.0 MYSAMPLE.RAW
```

You can also use an audio tool like Audacity to Export an audio sample to an uncompressed Stereo 22375 Hz, RAW (header-less) unsigned 8-bit PCM sample.

Audio Sample Handling (CoCo 3 only) - Overview

Use the SAMPLE_LOAD SAMPLE_LOAD fileName\$,Sample #

and SAMPLE PLAY #, SAMPLE LOOP # OR SAMPLE OFF command

```
' Handle SAMPLE command
' SAMPLE SINGLE 2 ' Play audio sample #2 one single time
' SAMPLE LOOP 4 ' Play audio sample #4 continuously looping
' SAMPLE OFF ' turns off any sample playing
```



```
ffmpeg -i source_audio.mp3 -acodec pcm_u8 -f u8 -ac 1 -ar 6003 -af
aresample=6003:filter_size=256:cutoff=1.0 MYSAMPLE.RAW
```

Sprite Handling - Overview

To use sprites in your BASIC programs you need to first prepare them with some external tools. - A graphic editor like GIMP to make or convert/export your sprite image into a 32 bit RGBA (includes transparency) PNG file. - The command line tool (included with this compiler) PNGtoCCSB to convert the PNG file into a Sprite to be used in your BASIC program.

Sprites are supported in every GMODE available in the compiler. Because each display mode has its own memory layout, palette, coordinate limits, and aspect ratio, use PNGtoCCSB to convert each sprite for the same GMODE that the BASIC program selects.

Typically you'll use PNGtoCCSB to convert your PNG file into a sprite with a command similar to:

```
PNGtoCCSB -g15 -s0 MySprite.PNG
PNGtoCCSB -g15 -s0 MyOtherSprite.PNG
```

This command will convert the PNG files MySprite.PNG & MyOtherSprite.PNG into compiled sprites ready to be used -g15 tells the tool we are going to use this sprite with a GMODE 15 graphics screen in your BASIC program. When converting the PNG to a compiled sprite it will match the colours of the PNG to the available colours in the GMODE & Screen mode (-sx option) on the CoCo.

This will save the compiled sprites as MySprite.asm & MyOtherSprite.asm

Character-cell sprites in GMODE 0 and GMODE 1

GMODE 0 and GMODE 1 are 32×16 character-cell screens. PNGtoCCSB treats every visible PNG pixel as one complete 8×12 display cell and expands the width by 3:2 so the sprite keeps the proportions of the source artwork. `SPRITE LOCATE` therefore uses native cell coordinates: X is 0–31 and Y is 0–15. These modes do not provide sub-cell movement.

```
PNGtoCCSB -g0 MyIASprite.PNG
PNGtoCCSB -g1 -s0 MyEASprite.PNG
```

GMODE 0 can reproduce opaque black plus all eight MC6847 semigraphics colours. GMODE 1 uses the portable full-cell bytes `$80`, `$BF`, and `$FF`: opaque black, foreground colour 1, and foreground colour 2. When a source image has three visible colours, PNGtoCCSB can represent the middle colour with a stable checker of the two foreground cells. Use `-s0` with `SCREEN 1,0`, or `-s1` with `SCREEN 1,1`.

On an original MC6847, the two GMODE 1 foregrounds are blue/red for colour set 0 or magenta/orange for colour set 1. On the MC6847T1 in a CoCo 2B, the same portable bytes remain

solid cells but appear as red/orange. Sprite shape, location, backup, drawing, erasing, and double buffering work on both video chips; only the GMODE 1 palette differs.

The converted width may not exceed 32 cells and the source image may not exceed 16 rows. Keep the complete sprite on screen: `X + sprite width <= 32` and `Y + sprite height <= 16`. There is no runtime edge clipping. As with other compiled sprites, reserve two pages by selecting page 1 once before returning to page 0, copy the completed background with `GCOPY 0,1`, and use `WAIT VBL` to process the sprite queue.

Semigraphics sprites in GMODE 2 through GMODE 8

PNGtoCCSB can generate sprites for every CoCo 1/2 semigraphics mode from GMODE 2 through GMODE 8. Select the exact mode used by the BASIC program:

```
PNGtoCCSB -g2 MySG4Sprite.PNG
PNGtoCCSB -g3 MySG4HSprite.PNG
PNGtoCCSB -g4 -s0 MySG6Sprite.PNG
PNGtoCCSB -g5 MySG6HSprite.PNG
PNGtoCCSB -g6 MySG8Sprite.PNG
PNGtoCCSB -g7 MySG12Sprite.PNG
PNGtoCCSB -g8 MySG24Sprite.PNG
```

GMODE 2 uses SG4 cells containing four logical pixels arranged as a 2×2 block. All four pixels in a byte share one colour. PNGtoCCSB maps each source pixel to a complete SG4 cell and expands the sprite horizontally by 3:2 to compensate for the physical shape of that cell. For example, an 8×8 PNG becomes 12 cells wide, or 24 logical pixels, and 16 logical pixels high. The aspect-corrected width may not exceed 32 cells and the source image may not exceed 16 rows.

GMODE 4 uses SG6 cells containing six logical pixels arranged as a 2×3 block. Each byte selects black plus one of two foreground colours. For flat artwork containing up to three visible source colours, PNGtoCCSB assigns distinct styles: solid foreground colour 1, a stable checker of foreground colours 1 and 2, and solid foreground colour 2. This keeps opaque regions filled and preserves details such as an outer shape, middle ring, and center even though SG6 cannot reproduce all three original hues. Artwork with more colours uses the closest of those three styles. The pattern is anchored to the animation frame, so it does not shimmer as the sprite moves. Use `-s0` for blue/violet and red/burgundy with `SCREEN 1,0`, or `-s1` for magenta and orange with `SCREEN 1,1`.

GMODE 4 requires the original MC6847 video chip. The MC6847T1 used by the CoCo 2B does not implement SG6 and interprets the same bytes as SG4, producing large incorrect blocks. In MAME, test GMODE 4 programs with the `coco` machine target rather than `coco2b`. The converter applies 3:2 horizontal aspect correction. An 8×8 PNG becomes 12 cells wide, or 24 logical pixels, and 24 logical pixels high. Because every generated sprite includes shifted variants and backs up one extra cell at its

right and bottom edges, the aspect-corrected width may not exceed 31 cells and the source image may not exceed 15 rows.

GMODE 3 and GMODE 5 through GMODE 8 are 64 hardware subpixels wide and store two horizontal subpixels in each screen byte. The two subpixels in a byte must share one colour. To reproduce the PNG colours faithfully, PNGtoCCSB maps each PNG pixel to one complete two-subpixel colour cell. A completely transparent PNG pixel leaves its cell untouched; a visible pixel draws and restores the complete cell. A sprite PNG may therefore be at most 32 pixels wide and its generated logical sprite width is twice the PNG width.

PNGtoCCSB automatically corrects the sprite height for the physical shape of a complete colour cell. GMODE 3 produces four logical rows for every three PNG rows, GMODE 5 produces two logical rows for every PNG row, GMODE 6 produces eight rows for every three PNG rows, GMODE 7 produces four rows for every PNG row, and GMODE 8 produces eight rows for every PNG row. GMODE 3 and GMODE 5 then duplicate each logical row into both physical memory rows required by their hybrid display modes. For example, an 8×8 PNG becomes an 11-logical-row GMODE 3 sprite stored in 22 physical rows. The corrected height must fit within the selected graphics screen.

Place GMODE 2 sprites at even X and Y coordinates for exact cell alignment. Place GMODE 4 sprites at an even X coordinate and a Y coordinate divisible by three. Other coordinates split source cells across adjacent bytes, so PNGtoCCSB selects the closest available shared colour for a split cell. In GMODE 3 and GMODE 5 through GMODE 8, only X needs to be even for exact colour reproduction.

At a shifted GMODE 4 edge, transparent subpixel bits are preserved, but all lit subpixels in an SG6 byte share the byte's single foreground-colour attribute. An underlying lit background pixel can therefore temporarily change hue where a sprite only partly covers its byte. This is an SG6 hardware limitation; cell-aligned placement avoids it.

For GMODE 2, the converter creates four drawing versions for every animation frame. GMODE 4 creates six versions for two X alignments and three Y alignments. The remaining semigraphics modes create two X-shifted versions. The generated assembly file records its required GMODE, and the compiler reports an error if it is loaded into a program using a different mode.

As with the other CoCo 1/2 sprite modes, reserve two graphics pages and keep the entire sprite inside the 64-pixel screen width.

In your BASIC program you will need to start your program with the following commands:

```
Sprite_Load "MySprite.asm", 0 ' Load sprite as #0
Sprite_Load "MyOtherSprite.asm", 1 ' Load sprite as #1
```

```
GMODE 15,1 'This will reserve graphics page 0 and graphics page 1
GMODE 15,0 'Set the current graphics page to use as 0
GCLS 0 'Colour the graphics screen colour 0
SCREEN 1,0 'Show the graphics screen (you could do this later)
```

Draw any background graphics for your program here...

Once your background graphics are drawn you need to copy screen 0 to screen 1 this is done with the GCOPY command. This is necessary as the sprite routines use double buffering where the screen is switched to show screen 1 the sprite updates are drawn on screen 0, once the sprites are updated, screen 0 is shown and screen 1 sprite updates are drawn.

GCOPY 0,1 'Copy graphics screen 0 to graphics screen 1

Now that your background is setup you are ready to draw the sprites on the screen using the following commands:

A normal Sprite usage routine would look like this:

```
SPRITE LOCATE 0, X1, Y1 ' Set the location of sprite 0
SPRITE BACKUP 0          ' Copy what's behind sprite 0
SPRITE SHOW 0, 0         ' Draw sprite 0, frame 0
SPRITE LOCATE 1, X2, Y2 ' Set the location of sprite 1
SPRITE BACKUP 1          ' Copy what's behind sprite 1
SPRITE SHOW 1, 0         ' Draw sprite 1, frame 0
```

Except for the SPRITE LOCATE command the above commands aren't executed at this point, they are added to a sprite command queue that is executed when the WAIT VBL command is used as:

```
WAIT VBL          ' This is when the actual sprite updates occur
```

Typically you will next want the sprite erase commands after the Wait VBL command, which again are only added to the sprite command queue at this point. It is usually best to reverse the order of the sprites for erasing.

```
SPRITE ERASE 1      ' Erase Sprite 1 (reverse order is good)
SPRITE ERASE 0      ' Erase Sprite 0
```

Typically you would have a routine to update your sprites like this:

```

SpriteUpdate:
SPRITE LOCATE 0, X1, Y1 ' Set the location of sprite 0
SPRITE BACKUP 0        ' Copy what's behind sprite 0
SPRITE SHOW 0, 0        ' Draw sprite 0, frame 0
SPRITE LOCATE 1, X2, Y2 ' Set the location of sprite 1
SPRITE BACKUP 1        ' Copy what's behind sprite 1
SPRITE SHOW 1, 0        ' Draw sprite 1, frame 0
WAIT VBL                ' This is when the actual sprite updates occur
SPRITE ERASE 1          ' Erase Sprite 1 (reverse order is good)
SPRITE ERASE 0          ' Erase Sprite 0
Return                  ' All done updating sprites

```

In your program you would change X1,Y1 & X2,Y2 to the locations you want and then update the sprites on the screen with the command:

```
GOSUB SpriteUpdate
```

The last thing to know about the sprites is they can also have a frame option. If you create or download a sprite sheet of an animated sprite, they are usually created as many images in a row. For example a sprite sheet might look something like this:



Illustration from the original BasTo6809 manual.

```

If you wanted to use this sprite in your program you would use the -axx option with the
PNGtoCCSB tool in this case we have 8 frames of animation so we would use the following
command:
PNGtoCCSB -g15 -s0 -a8 WolfCub.PNG

```

The -a8 tells the tool to create a compiled sprite with 8 frames from the one PNG file. The PNG must only have the images in one horizontal row and be evenly spaced.

In your program you use the same SPRITE commands as above except you also now want to use the frame option of the SPRITE SHOW command where the 8 frames to show are from frame 0 to frame 7

```
Sprite_Load "WolfCub.asm", 2 ' Load sprite as #2
```

The sprite update code would now look like this:

```
SpriteUpdate:
SPRITE LOCATE 0, X1, Y1 ' Set the location of sprite 0
SPRITE BACKUP 0      ' Copy what's behind sprite 0
SPRITE SHOW 0, 0      ' Draw sprite 0, frame 0
SPRITE LOCATE 1, X2, Y2 ' Set the location of sprite 1
SPRITE BACKUP 1      ' Copy what's behind sprite 1
SPRITE SHOW 1, 0      ' Draw sprite 1, frame 0
SPRITE LOCATE 2, WolfCubX, WolfCubY ' Set the location of sprite 2
SPRITE BACKUP 2      ' Copy what's behind sprite 2
SPRITE SHOW 2, F      ' Draw sprite 2, frame F
WAIT VBL              ' This is when the actual sprite updates occur
SPRITE ERASE 2        ' Erase Sprite 2 (reverse order is good)
SPRITE ERASE 1        ' Erase Sprite 1
SPRITE ERASE 0        ' Erase Sprite 0
Return                ' All done updating sprites
```

Sprites are supported in every GMODE available in the compiler. GMODE 0 and GMODE 1 use the whole-cell rules above; GMODE 2 through GMODE 8 use the semigraphics cell rules above; the remaining GMODEs use their packed-pixel sprite formats.

CoCo 1 & 2 Graphic Modes - GMODE

GMODE #	Resolution	Colours	Bytes Per Screen	Mode Name
0	32 x 16	9	512	Internal alphanumeric
1	32 x 16	2	512	External alphanumeric
2	64 x 32	9	512	Semi graphic-4
3	64 x 32	9	2048	Semi graphic-8*
4	64 x 48	9**	512	Semi graphic-6
5	64 x 48	9	3072	Semi graphic-12***
6	64 x 64	9	2048	Semi graphic-8
7	64 x 96	9	3072	Semi graphic-12
8	64 x 192	9	6144	Semi graphic-24
9	64 x 64	4	1024	Full graphic 1-C
10	128 x 64	2	1024	Full graphic 1-R
11	128 x 64	4	2048	Full graphic 2-C
12	128 x 96	2	1536	Full graphic 2-R
13	128 x 96	4	3072	Full graphic 3-C
14	128 x 192	2	3072	Full graphic 3-R
15	128 x 192	4	6144	Full graphic 6-C
16	256 x 192	2	6144	Full graphic 6-R

GMODE #	Resolution	Colours	Bytes Per Screen	Mode Name
17			6144	DMA Mode (unusable)
18	256 x 192	2	6144	Compile for Artifacts

- Semi graphics 4 Hybrid using Semi Graphics 8 Mode ** 3 Colours for each colour set *** Semi graphics-6 Hybrid using Semi Graphics 12 mode

CoCo 3 Graphic Modes - GMODE

GMODE #	Resolution	Colours	Bytes Per Screen
100	64 x 192	4	3200
101	64 x 200	4	3200
102	64 x 225	4	3600
103	64 x 192	16	6144
104	64 x 200	16	6400
105	64 x 225	16	7200
106	80 x 192	4	3840
107	80 x 200	4	4000
108	80 x 225	4	4500
109	80 x 192	16	7680
110	80 x 200	16	8000
111	80 x 225	16	9000
112	128 x 192	2	3072
113	128 x 200	2	3200
114	128 x 225	2	3600
115	128 x 192	4	6144
116	128 x 200	4	6400

GMODE #	Resolution	Colours	Bytes Per Screen
117	128 x 225	4	7200
118	128 x 192	16	12288
119	128 x 200	16	12800
120	128 x 225	16	14400
121	160 x 192*	2	3840
*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen	*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen	*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen	*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen
122	160 x 200*	2	4000
*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen	*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen	*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen	*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen
123	160 x 225*	2	4500
*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen	*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen	*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen	*(viewable) really 128x192, * Special mode that repeats the left 4 bytes on the right side of the screen
124	160 x 192	4	7680

GMODE #	Resolution	Colours	Bytes Per Screen
125	160 x 200	4	8000
126	160 x 225	4	9000
127	160 x 192	16	15360
128	160 x 200	16	16000
129	160 x 225	16	18000
130	256 x 192	2	6144
131	256 x 200	2	6400
132	256 x 225	2	7200
133	256 x 192	4	12288
134	256 x 200	4	12800
135	256 x 225	4	14400
136	256 x 192	16	24576
137	256 x 200	16	25600
138	256 x 225	16	28800
139	320 x 192	2	7680
140	320 x 200	2	8000
141	320 x 225	2	9000
142	320 x 192	4	15360

GMode #	Resolution	Colours	Bytes Per Screen
143	320 x 200	4	16000
144	320 x 225	4	18000
145	320 x 192	16	30720
146	320 x 200	16	32000
147	320 x 225	16	36000
148	512 x 192	2	12288
149	512 x 200	2	12800
150	512 x 225	2	14400
151	512 x 192	4	24576
152	512 x 200	4	25600
153	512 x 225	4	28800
154	640 x 192	2	15360
155	640 x 200	2	16000
156	640 x 225	2	18000
157	640 x 192	4	30720
158	640 x 200	4	32000
159	640 x 225	4	36000
160*	128 x 192	256	24576

GMODE #	Resolution	Colours	Bytes Per Screen
161*	128 x 200	256	25600
162*	128 x 225	256	28800
163*	160 x 192	256	30720
164*	160 x 200	256	32000
165*	160 x 225	256	36000

- These modes are NTSC composite CoCo 3 output only

Specifications

SDC_BIGLOADM"FILENAME.BIN",# Loads CoCo 3 Memory Blocks very fast. Useful for loading in background screens for games or other large amounts of data. # is the file number 0 or 1

This command loads complete \$2000 byte blocks into the CoCo 3's memory. The format of this file is written in 512 byte chunks, as the SDC streaming mode reads file data 512 bytes at a time. Chunk 0 is the header/Memory Mapping Chunk:

Bytes

0 & 1 File version #

Version 1 is the only supported version to date

2 & 3 Is the 16 bit number of the first bank to load

x & x+1 Is the 16 bit number of the next bank to load and so on until we get a \$FFFF

If we reach a \$FFFF this signifies we've copied the last block



and are done.

512 & 513 Chunk 1 - This is the actual first word of data for the first block

This contains \$2000 bytes of data, the next \$2000 bytes will be

the data for the 2nd bank number given in the memory mapping Chunk and will continue with \$2000 bytes of data for

all the rest of the banks given in the memory map Chunk.

Example Programs

This is a tweaked version of James Diffendaffer's 3D plot program that I converted from working on a CoCo 3 to work on a CoCo 1 & 2

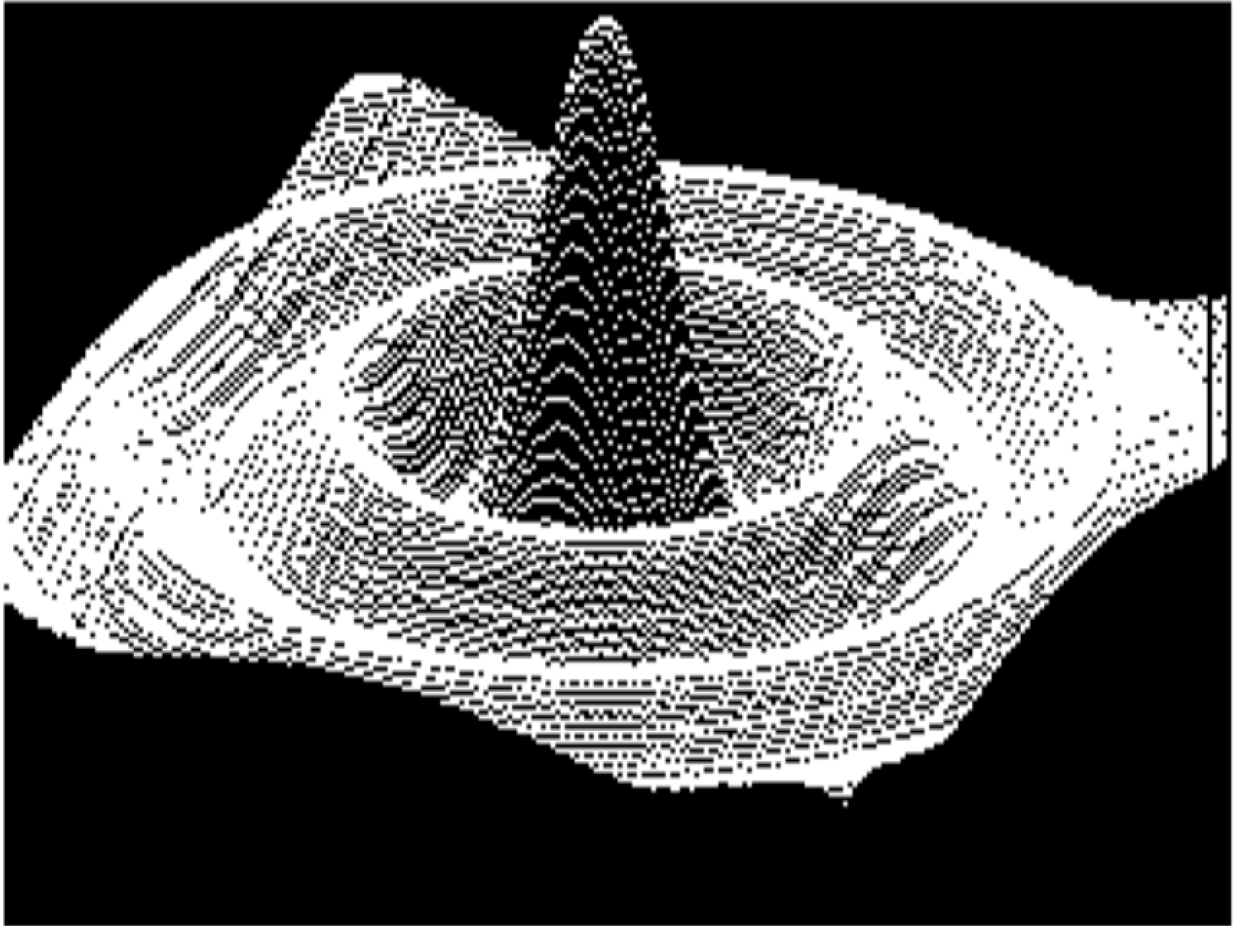


Illustration from the original BasTo6809 manual.

Original program (EXTENDED COLOR BASIC):

```

0 CX=255:CY=192:PMODE 4,1:PCLS:SCREEN 1,1
1 DIM R(255):FOR I=0 TO CX:R(I)=CY:NEXT I:GOTO 10
2 R=SQR(X*X+Y*Y)*1.5: IF R=0 THEN F=90:GOTO 4
3 F=90*SIN(R)/R
4 A=10*X+125-5*Y:B=5*Y+2.5*X+93:RETURN
10 FOR Y=10 TO -10 STEP -0.1
70 FOR X=10 TO -10 STEP -0.1
80 GOSUB 2
82 IF A<0 THEN A=0
83 IF A>255 THEN A=255
84 IF R(A)>B-F THEN R(A)=B-F:PSET(A,B-F)
90 NEXT X,Y
101 GOTO 101

```

Below is the same program modified to work with the compiler. The only differences are: (The SDECB IDE auto indents the code) 1) The variables are declared with the DIM command 2) GMODE instead of PMODE 3) GCLS instead of PCLS 4) Line 84 the SET command is used which includes a colour value, instead of PSET

```

' compileoptions -s20
' Declare the format Type of numeric variables
DIM CX AS _UNSIGNED _BYTE, CY AS _UNSIGNED _BYTE
DIM R(255) AS SINGLE
DIM A AS SINGLE, B AS SINGLE
DIM F AS SINGLE
DIM R AS SINGLE
DIM X AS SINGLE, Y AS SINGLE
DIM I AS _UNSIGNED _BYTE
0 CX = 255: CY = 192: GMODE 16, 0: GCLS: SCREEN 1, 1
1 FOR I = 0 TO CX: R(I) = CY: NEXT I: GOTO 10
2 R = SQR(X * X + Y * Y) * 1.5: IF R = 0 THEN F = 90: GOTO 4
3 F = 90 * SIN(R) / R
4 A = 10 * X + 125 - 5 * Y: B = 5 * Y + 2.5 * X + 93: RETURN
10 FOR Y = 10 TO -10 STEP -0.1
70 FOR X = 10 TO -10 STEP -0.1
80 GOSUB 2
82 IF A < 0 THEN A = 0
83 IF A > 255 THEN A = 255
84 IF R(A) > B - F THEN R(A) = B - F: SET (A, B - F, 1)
90 NEXT X, Y
101 GOTO 101

```

This is a sample of using a 640x225, 4 colour mode on a CoCo 3 Thanks to bluetip on the CoCo Nation Discord server



Illustration from the original BasTo6809 manual.

```
' compileoptions -s20
' Declare the format of numeric variables
DIM XMAX AS _UNSIGNED INTEGER
DIM X AS _UNSIGNED INTEGER
DIM YMAX AS _UNSIGNED _BYTE
DIM Y AS _UNSIGNED _BYTE
DIM Z AS _UNSIGNED INTEGER
GMODE 159
PALETTE 0, 0
PALETTE 1, 63
PALETTE 2, 18
PALETTE 3, 36
GCLS 0
SCREEN 1, 0
```

XMAX = 639

YMAX = 224

```

FOR Z = 0 TO 32767
SET (RND(XMAX), RND(YMAX), RND(2) + 1)
NEXT Z
FOR Y = 0 TO YMAX STEP 8
LINE (0, Y)-(XMAX, Y), 1
NEXT Y
FOR X = 0 TO XMAX STEP 8
LINE (X, 0)-(X, YMAX), 1
NEXT X
Finish:
GOTO Finish

```

TEST.BAS

```

CoCo 1/2 Hi-res dots
' compileoptions -s20
DIM X AS _UNSIGNED INTEGER
DIM I AS _UNSIGNED _BYTE
10 CLS: PRINT "WELCOME TO THE COCO COMPILER    TEST PROGRAM. IF YOU ARE READINGTHIS THAN
YOU HAVE SUCCESSFULLY COMPILED THE TEST PROGRAM."
20 PRINT: INPUT "PRESS ENTER TO SEE THE FG6R    SCREEN FILL WITH 20,000 DOTS"; A$
30 GMODE 16, 0: GCLS: SCREEN 1, 1
40 FOR X = 1 TO 20000
50 SET (RND(256) - 1, RND(192~%%) - 1, 1)
60 NEXT X
70 SOUND 100, 20
80 SCREEN 0, 0
90 CLS
100 PRINT: PRINT " ALL GOOD, NOW COMPILE YOUR OWN    BASIC PROGRAMS AND HAVE FUN"
110 PRINT: PRINT: PRINT "    COCO FOREVER!!!"
120 FOR I = 1 TO 200 STEP 25
130 SOUND I, 1
140 NEXT I

```

GETFSIZE.BAS

Get the size of a file stored on the actual SD card's filesystem on the CoCoSDC (not the files in a .DSK image)

```

' compileoptions -s20
' Print length of a file on the CoCo SDC
DIM FileLength AS _UNSIGNED LONG ' LONG is a 32 bit number
CLS
PRINT "ENTER THE FILENAME TO GET"
INPUT "THE SIZE OF"; Filename$
SDC_OPEN Filename$, "R", 1 ' Open file for reading

```

A\$ = SDC_FILEINFO\$(1) ' Get fileinfo in A\$

```
SDC_CLOSE (1) 'close file
PRINT: PRINT "RAW HEX CODES OF FILE INFO:"
FOR I = 1 TO LEN(A$)
PRINT RIGHT$("0" + HEX$(ASC(MID$(A$, I, 1))), 2);
NEXT I
PRINT
' Size is stored LSB first
```

FileLength = ASC(MID\$(A\$, 29, 1)) + ASC(MID\$(A\$, 30, 1)) * 256 + ASC(MID\$(A\$, 31, 1)) * 65536 +
ASC(MID\$(A\$, 32, 1)) * 16777216

```
PRINT "LENGTH OF FILE: "; Filename$
PRINT "IS: "; FileLength; " BYTES"
Finished:
GOTO Finished
```

BASASSEM.BAS

Interface BASIC with assembly code

```

' compileoptions -s20
' Example how to interface BASIC and assembly code
DIM N AS INTEGER
5 CLS: PRINT @32,
10 INPUT "ENTER A NUMBER FROM -32000 TO 32000"; N
20 PRINT N; "+ 7 IS";
30 GOSUB 1000 ' Goto assembly code and do an add
35 PRINT N
40 INPUT "WHAT IS YOUR NAME"; Name$
50 GOSUB LearnStrings ' Copy the Name$ string to the top of the screen
60 PRINT "NICE NAME "; Name$
70 END
1000 ' Example of some assembly code
' ADDASSEM:
; Comments in assembly code must start with a semicolon or an asterisks
; Comment is BASIC code must use apostrophe or REM
; Let's access the BASIC numeric variable N
; and multiply it by 2
    LDD    _Var_N ; Get the signed 16 variable N into D
    ADDD   #$0007 ; Add 7
    STD    _Var_N ; Store the result back into variable N
; Can have the RTS here or use RETURN just past the ENDASSEM:
' ENDASSEM:
RETURN
LearnStrings:
' ADDASSEM:
; Example of how to access a string variable in assembly code
; Copy the string to the top of the text screen
    LDB    _StrVar_Name ; Get the length of the string
    LDU    #_StrVar_Name+1 ; Point at the actual string data
    LDX    #$400 ; Point at the top left corner of the text screen
!   LDA    ,U+ ; Get the next character
    STA    ,X+ ; Store the character on the screen
    DECB   ; Decrement the counter
    BNE    < ; Loop until counter is 0
    RTS
' ENDASSEM:
' The compiler adds a prefix to variables (which you can see at the beginning of the
.asm file it genereates)
' numeric variables are prefixed with _Var_ - numeric variables are signed 16 bit
numbers
' string variables are prefixed with _StrVar_ - strings are stored beginning at the
address of the variable
' the format is - byte 0 is the length of the string, the actual string starts at byte 1
,

```

PAGEFLIP.BAS

Example of GMODE, GCOPY, CIRCLE and PAINT commands to do simple animation

```
' compileoptions -s16
' Example of doing page animation
' Must always setup the number of pages you will use as the
' first GMODE command in your program
' With this many pages you will require 64k and will need to use
' the cc1sl (CoCo Super Loader program) see the Manual.pdf
DIM S AS _BYTE ' Range of -128 to 127
DIM X AS _UNSIGNED_BYTE ' Range from 0 to 255
DIM Frames AS _UNSIGNED_BYTE ' RANGE 0 to 255
' First GMODE sets graphic mode and reserves 7 pages 0 to 6
GMODE 16, 6
GMODE , 0 ' Set the active graphics page to 0
GCLS 0 ' Colour the screen colour 0 which is black
```

Frames = 6 ' Number of frames to use

```
FOR X = 1 TO 2000 ' Draw some dots on the screen
SET (RND(256) - 1, RND(192) - 1, 1)
NEXT X
FOR X = 1 TO Frames ' Copy page 0 to all the other pages
40 GCOPY 0, X
NEXT X
' Draw and fill a circle on the screen
50 X = 20
FOR S = 0 TO Frames
GMODE , S
CIRCLE (X, 20), 11, 1
PAINT (X, 20), 0, 1
```

X = X + 10

```
NEXT S
' Ping Pong animation frames (Loops forever)
100 FOR S = 0 TO Frames
GMODE , S: SCREEN 1, 1 ' Screen shows the selected page
SOUND 10, 5
NEXT S
FOR S = Frames - 1 TO 1 STEP -1
GMODE , S: SCREEN 1, 1 ' Screen shows the selected page
SOUND 10, 5
NEXT S
GOTO 100
```

SDCDIR.BAS

Show a directory of the files on the CoCoSDC FAT32 filesystem

```
' compileoptions -s20
DIM X AS _BYTE
DIM Y AS _UNSIGNED _BYTE
CLS: PRINT "SDC ACCESS"
```

CurrentDir\$ = "." ' Start at the top folder on the SDC

```
' CurrentDir$="TEST1/*.*" ' Start at the folder TEST on the SDC
PRINT "PATH: "; CurrentDir$
' Set the initial directory location
```

X = SDC_INITDIR(CurrentDir\$): GOSUB CheckForError

```
PRINT "DIRECTORY:"
NextPage:
```

X = SDC_DIRPAGE(0): GOSUB CheckForError ' Get 16 directory entries

Y = 0

```
WHILE Y < 16
```

A\$ = SDC_DIRLIST\$(Y) ' A\$ = The 16 byte entry

```
IF ASC(LEFT$(A$, 1)) = 0 THEN END ' If entry is blank then we've reached the last entry
FOR X = 1 TO 8: PRINT MID$(A$, X, 1);: NEXT X
PRINT ".";
FOR X = 9 TO 11: PRINT MID$(A$, X, 1);: NEXT X
PRINT
```

Y = Y + 1

```
WEND
GOTO NextPage
CheckForError:
IF X <> 0 THEN PRINT "ERROR NUMBER IS: "; X: END
RETURN
```

CONSTANT.BAS

Show an example of using constants in your program.

```
' compileoptions -s20
' Declare a numeric constant approximately equal to the ratio of a circle's
' circumference to its diameter:
Const PI = 3.141593
' Declare some string constants:
Const circumferenceText = "THE CIRCUMFERENCE OF THE CIRCLE IS"
Const areaText = "THE AREA OF THE CIRCLE IS"
Do
Input "ENTER THE RADIUS OF A CIRCLE OR ZERO TO QUIT"; radius
If radius = 0 Then End
Print circumferenceText; 2 * PI * radius
Print areaText; PI * radius * radius ' radius squared
Print
Loop
End
```



Thanks

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Thanks to bluetip